IT Specialist

Device Configuration and Management

Student Edition

Presented By

LearnKey®

IT Specialist – Device Configuration and Management Project Workbook

First Edition

LearnKey creates signature multimedia courseware. LearnKey provides expert instruction for popular computer software, technical certifications, and application development with dynamic video-based courseware and effective learning management systems. For a complete list of courses, visit https://www.learnkey.com.

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Table of Contents

Introduction **1**

Best Practices Using LearnKey's Online Training 2

Using This Workbook 3

Skills Assessment 4

ITS – Device Configuration and Management Video Times 6

Domain 1 Lesson 1 **7**

RAM 8

CPU, APU, ARM 9

Graphics Card 10

Physical Storage 11

Motherboard 12

Domain 1 Lesson 2 **13**

Domain User Account 14

Domain Administrator Account 15

Centralized Management 16

Group Policy 17

Local and Group Security Policies 18

Domain 1 Lesson 3 **19**

Installation and Account Options 20

Domain and Non-Domain Devices 21

Domain 1 Lesson 4 **22**

User Accounts Types 23

User and Administrative Accounts 24

User Profiles 25

Domain 1 Lesson 5 **26**

Start Menu 27

Display Settings 28

Application Shortcuts 29

Time Zone Settings 30

Domain 1 Lesson 6 **31**

Taskbar Personalization 32

Power Settings 33

Window Management 34
Domain 1 Lesson 7 **35**
Mouse Settings 36
Color Filters and Contrast Themes 37
Audio Settings 38
Magnifier 39
Narrator 40
Domain 1 Lesson 8 **41**
Sticky Keys 42
On-Screen Keyboard 43
Voice Access and Voice Typing 44
Text Cursor and Eye Control 45
Live Captions 46
Domain 1 Lesson 9 **47**
Windows Update Settings 48
Software Updates and Patches 49
Optional OS Updates 50
Device Driver Updates 51
Update History 52
Domain 2 Lesson 1 **53**
Install Applications 54
Modify Applications 55
Remove Desktop Applications 56
Domain 2 Lesson 2 **57**
Optional Features 58
The Microsoft Store 59
Application Storage 60
Remote Desktop 61
Domain 2 Lesson 3 **62**
Using Copilot 63
Copilot Limitations 64
Domain 2 Lesson 4 **65**
The Windows Registry 66
HDMI 67

DisplayPort 68
DVI 69
Domain 2 Lesson 5 **70**
VGA 71
USB-A, Mini-A, Micro-A, and USB 3.0 72
USB-B, Mini-B, Micro-B, and USB 3.0 73
USB-C 74
Thunderbolt 75
Domain 2 Lesson 6 **76**
S/PDIF 77
AUX Cable 78
HDMI Converters 79
Domain 2 Lesson 7 **80**
Wireless Casting 81
Display Orientation 82
Duplicating and Extending 83
Resolution and Aspect Ratio 84
ClearType 85
Domain 3 Lesson 1 **86**
Cloud Services 87
Storage and Service Providers 88
File Sharing and Synchronization 89
Domain 3 Lesson 2 **90**
Partition and Format a Drive 91
Files and Folders 92
GPT and MBR Partition Styles 93
Domain 3 Lesson 3 **94**
File, Share, and Effective Permissions 95
Basic and Advanced Permissions 96
Mapping a Drive and Taking Ownership 97
Domain 3 Lesson 4 **98**
Backup and Restore 99
Domain 4 Lesson 1 **100**
Allow an App or Feature 101

Network Profiles 102
Firewall Troubleshooting 103
Domain 4 Lesson 2 104
Multifactor Authentication 105
Biometric Authentication 106
Windows Sign-In Options 107
Strong Passwords 108
Authenticator Apps 109
Domain 4 Lesson 3 110
Computer Viruses 111
Adware 112
Spyware 113
Denial of Service (DoS) Attacks 114
Social Engineering Attacks 115
Physical Attacks 116
Domain 4 Lesson 4 117
Antimalware App Options 118
Antimalware Scan Results 119
Domain 4 Lesson 5 120
User Account Control (UAC) 121
Domain 5 Lesson 1 122
Windows Troubleshooting Tools 123
Gather Data on Issues 124
Research Issue Remedies 125
Identify When to Escalate Issues 126
Force Group Policy Application 127
Problems with Applied Policies 128
Domain 5 Lesson 2 129
Resets and Rollbacks 130
Advanced Startup 131
Safe Mode 132
Application Compatibility Issues 133
Domain 5 Lesson 3 134
Microsoft Store Application Issues 135

Repair or Reinstall an Application 136
S Mode 137
Troubleshoot Windows Services 138
Task Manager 139
Domain 5 Lesson 4 140
Troubleshoot Hardware 141
Update or Roll Back Drivers 142
Uninstall and Reinstall Drivers 143
Device Manager and Disk Management 144
Peripheral Devices 145
Domain 5 Lesson 5 146
Network Connectivity Issues 147
Join or Remove Devices from a Domain 148
Appendix 149
Glossary 150
Objectives 155
IT Specialist – Device Configuration and Management Lesson Plan 158
Domain 1 Lesson Plan 159
Domain 2 Lesson Plan 161
Domain 3 Lesson Plan 163
Domain 4 Lesson Plan 164
Domain 5 Lesson Plan 166

Introduction

IT Specialist

Device Configuration and Management

Best Practices Using LearnKey's Online Training

LearnKey offers video-based training solutions that are flexible enough to accommodate private students and educational facilities and organizations.

Our course content is presented by top experts in their respective fields and provides clear and comprehensive information. The full line of LearnKey products has been extensively reviewed to meet superior quality standards. Our course content has also been endorsed by organizations such as Certiport, CompTIA®, Cisco, and Microsoft. However, it is the testimonials given by countless satisfied customers that truly set us apart as leaders in the information training world.

LearnKey experts are highly qualified professionals who offer years of job and project experience in their subjects. Each expert has been certified at the highest level available for their field of expertise. This expertise provides the student with the knowledge necessary to obtain top-level certifications in their chosen field.

Our accomplished instructors have a rich understanding of the content they present. Effective teaching encompasses presenting the basic principles of a subject and understanding and appreciating organization, real-world application, and links to other related disciplines. Each instructor represents the collective wisdom of their field and within our industry.

Our Instructional Technology

Each course is independently created based on the manufacturer's standard objectives for which the course was developed.

We ensure that the subject matter is up-to-date and relevant. We examine the needs of each student and create training that is both interesting and effective. LearnKey training provides auditory, visual, and kinesthetic learning materials to fit diverse learning styles.

Course Training Model

The course training model allows students to undergo basic training, building upon primary knowledge and concepts to more advanced application and implementation. In this method, students will use the following toolset:

Pre-assessment: The pre-assessment is used to determine the student's prior knowledge of the subject matter. It will also identify a student's strengths and weaknesses, allowing them to focus on the specific subject matter they need to improve the most. Students should not necessarily expect a passing score on the pre-assessment as it is a test of prior knowledge.

Video training sessions: Each training course is divided into sessions or domains and lessons with topics and subtopics. LearnKey recommends incorporating all available external resources into your training, such as student workbooks, glossaries, course support files, and additional customized instructional material. These resources are located in the folder icon at the top of the page.

Exercise labs: Labs are interactive activities that simulate situations presented in the training videos. Step-by-step instructions and live demonstrations are provided.

Post-assessment: The post-assessment is used to determine the student's knowledge gained from interacting with the training. In taking the post-assessment, students should not consult the training or any other materials. A passing score is 80 percent or higher. If the individual does not pass the post-assessment the first time, LearnKey recommends incorporating external resources, such as the workbook and additional customized instructional material.

Workbook: The workbook has various activities, including fill-in-the-blank questions, short answer questions, practice exam questions, and group and individual projects that allow the student to study and apply concepts presented in the course videos.

Using This Workbook

This project workbook contains practice projects and exercises to reinforce the knowledge you have gained through the video portion of the **IT Specialist – Device Configuration and Management** course. The purpose of this workbook is twofold. First, get you further prepared to pass the IT Specialist – Device Configuration and Management exam, and second, to teach you job-ready skills and increase your employability in the area of Windows device support.

The projects within this workbook follow the order of the video portion of this course. To save your answers in this workbook, you must first download a copy to your computer. You will not be able to save your answers in the web version. You can complete the workbook exercises as you go through each section of the course, complete several at the end of each domain, or complete them after viewing the entire course. The key is to go through these projects to strengthen your knowledge in this subject.

Each project is based upon a specific video (or videos) in the course and specific test objectives. The materials you will need for this course include:

- LearnKey's **IT Specialist – Device Configuration and Management** courseware.
- A device with Windows 11 installed.
- The course project files. All applicable project files are located in the support area where you downloaded this workbook.
- Note: This workbook follows the Windows 11 version of this IT Specialist – Device Configuration and Management course.

For Teachers

LearnKey is proud to provide extra support to instructors upon request. For your benefit as an instructor, we also provide an instructor support .zip file containing answer keys, completed versions of the workbook project files, and other teacher resources. This .zip file is available within your learning platform's admin portal.

Notes

- Extra teacher notes, when applicable, are in the Project Details box within each exercise.
- Exam objectives are aligned with the course objectives listed in each project, and project file names correspond with these numbers.
- The Finished folder in each domain has reference versions of each project. These can help you grade projects.
- Short answers may vary but should be similar to those provided in this workbook.
- Teachers may consider asking students to add their initials, student ID, or other personal identifiers at the end of each saved project.
- Refer to your course representatives for further support.

We value your feedback about our courses. If you have any questions, comments, or concerns, please let us know by visiting https://about.learnkey.com.

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Skills Assessment

Instructions: Rate your skills on the following tasks from 1-5 (1 being needs improvement, 5 being excellent).

Skills	1	2	3	4	5
Recognize basic computer components and explain their functions.	☐	☐	☐	☐	☐
Describe the purpose of Active Directory services.	☐	☐	☐	☐	☐
Install Windows using the default settings.	☐	☐	☐	☐	☐
Configure user account options.	☐	☐	☐	☐	☐
Configure desktop settings.	☐	☐	☐	☐	☐
Configure accessibility settings to meet users' specific needs.	☐	☐	☐	☐	☐
Manage updates.	☐	☐	☐	☐	☐
Manage applications and Windows features.	☐	☐	☐	☐	☐
Explain the purpose and capabilities of Windows Copilot.	☐	☐	☐	☐	☐
Describe the purpose of the Windows Registry.	☐	☐	☐	☐	☐
Compare and contrast capabilities of peripheral connection types.	☐	☐	☐	☐	☐
Configure projection and display properties.	☐	☐	☐	☐	☐
Describe cloud services.	☐	☐	☐	☐	☐
Describe and configure storage.	☐	☐	☐	☐	☐
Describe and configure local and network file sharing and permissions.	☐	☐	☐	☐	☐
Manage backup and restore of user files and states.	☐	☐	☐	☐	☐
Configure Windows Defender Firewall Settings.	☐	☐	☐	☐	☐
Describe user authentication and configure Windows sign-in options.	☐	☐	☐	☐	☐
Describe various security threats.	☐	☐	☐	☐	☐
Describe how to respond to various malware and social engineering attacks.	☐	☐	☐	☐	☐

Skills	1	2	3	4	5
Manage User Account Control (UAC) settings.	☐	☐	☐	☐	☐
Perform troubleshooting tasks.	☐	☐	☐	☐	☐
Troubleshoot operating system and application issues.	☐	☐	☐	☐	☐
Manage and troubleshoot hardware and peripherals.	☐	☐	☐	☐	☐
Manage and troubleshoot device connections to networks and domains.	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐

ITS – Device Configuration and Management Video Times

Domain 1	Video Time
Basic Computer Components	00:17:50
Active Directory Services	00:12:20
Install Windows	00:06:08
User Account Options	00:09:23
Desktop Settings	00:16:46
Taskbar and Power Settings	00:10:39
Accessibility Settings	00:14:08
Keyboard, Voice, and Cursor Settings	00:13:06
OS and App Updates	00:11:14
Total Time	**01:51:34**

Domain 2	Video Time
Manage Apps	00:09:54
Manage Windows Features	00:08:50
Windows Copilot	00:10:06
Windows Registry and Video Connection Types	00:11:41
VGA, USB, and Thunderbolt	00:08:31
Audio and Conversion Connectors	00:06:57
Projection and Display Properties	00:11:17
Total Time	**01:07:16**

Domain 3	Video Time
Cloud Services	00:08:23
Storage	00:06:52
Local and Network File Permissions	00:07:55
Backup and Restore	00:04:55
Total Time	**00:28:05**

Domain 4	Video Time
Windows Defender Firewall	00:05:20
User Authentication Options	00:08:49
Various Security Threats	00:15:31
Antimalware Options and Results	00:07:52
UAC Settings	00:10:43
Total Time	**00:48:15**

Domain 5	Video Time
Troubleshooting Tasks	00:17:47
Operating Systems and Applications	00:14:27
Troubleshooting Apps	00:16:13
Hardware and Peripherals	00:12:01
Network and Domain Connections	00:08:12
Total Time	**00:48:27**

Domain 1

Lesson 1

IT Specialist

Device Configuration and Management

RAM

Random access memory (RAM) is a critical component of computing devices. It serves as a system's short-term memory and enables quick access to data. RAM is a form of volatile memory, meaning it loses stored data when power is turned off. It temporarily stores data that the central processing unit (CPU) needs to access quickly and is inserted into the motherboard's RAM slots.

Purpose

Upon completing this project, you will better understand RAM's hardware and software aspects, its different types, and how it impacts system performance.

Steps for Completion

1. Label the generations of DDR RAM sticks.

   a. DDR2 

   b. DDR 3 

   c. DDR4 DIMM stick 

2. Label the following statements as true or false.

   a. true Dynamic RAM (DRAM) requires constant refreshing to maintain data.

   b. false Static RAM (SRAM) is used for cache memory and is the least expensive.

   c. true RAM sticks come in different sizes and pin configurations.

   d. false Applications, like web browsers and video editing software, do not consume much RAM.

   e. true Factors that affect RAM performance include capacity, speed, and latency.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Basic Computer Components
Subtopic: RAM

Objectives covered
1 Install and Configure Windows
1.1 Recognize basic computer components and explain their functions
1.1.1 RAM

CPU, APU, ARM

Central processing units (CPUs) and accelerated processing units (APUs) handle calculations, execute instructions, and manage various processes within computing devices. A CPU is responsible for executing instructions from software and the operating system. An APU combines CPU and graphics processing unit (GPU) functionality into a single chip.

Purpose

Upon completing this project, you will better understand CPU architectures.

Steps for Completion

1. Open your device's Task Manager.
2. Observe your device's CPU before and after opening a browser.
3. CPUs with x86 architecture are often used in desktop and server environments.
4. CPUs with ARM are optimized for energy efficiency and are commonly found in mobile devices, tablets, and embedded systems.
5. APU are particularly useful in budget-friendly systems and compact devices.
6. List at least three factors that influence CPU and APU performance.
   a. - Instruction set compatibility
      - Power consomption
      - Cooling solutions

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 1
Topic: Basic Computer Components
Subtopic: CPU, APU, ARM

Objectives covered
1 Install and Configure Windows
1.1 Recognize basic computer components and explain their functions
1.1.2 CPU/APU (x86 [32-bit and 64-bit], ARM)

Graphics Card

Graphics cards contain graphics processing units (GPUs). GPUs are specialized processors that accelerate graphics rendering and computational workloads. They interact with software by accelerating image rendering and providing real-time computation for applications. The two main types of GPUs are integrated and dedicated. Integrated GPUs are built into the CPU and share system memory, while dedicated GPUs have their own memory and processing power.

Purpose

Upon completing this project, you will better understand the types of GPUs, their functions, key specifications, and how they impact system performance.

Steps for Completion

1. Open your device's Task Manager.
2. Observe your device's GPU performance.
3. List the four elements of a graphics card.
    a. GPU
    b. VRAM (Video RAM)
    c. Cooling solutions
    d. Various output interfaces
4. List at least three factors that influence GPU performance.
    a. the number of CUDA cores for (NVIDIA) or stream processors for (AMD), memory bandwidth, and thermal efficiency

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Basic Computer Components
Subtopic: Graphics Card

Objectives covered
1 Install and Configure Windows
1.1 Recognize basic computer components and explain their functions
1.1.3 Graphics card

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Physical Storage

Physical storage devices are responsible for storing and retrieving data within computing systems. Storage interacts with software by facilitating data retrieval, operating system boot times, and application performance. Factors that influence storage performance include read/write speeds, endurance, and connectivity type.

Purpose

Upon completing this project, you will better understand different types of physical storage, their characteristics, and how they impact system performance.

Steps for Completion

1. Open your device's Disk Management.
2. Review the disks and drives on your device and their key specifications.
3. How many lettered drives do you have?
   a. 1
4. Match the physical storage types to their descriptions.

A. Hard disk drives (HDDs)	C. Hybrid storage solutions
B. Solid state drives (SSDs)	

   a. A This storage uses spinning magnetic disks to read and write data, making them cost-effective for large storage capacities but slower than other options.
   b. C This storage combines other storage capacity with speed, balancing performance and storage space.
   c. B This storage utilizes flash memory to provide significantly faster read and write speeds, improving system responsiveness and reducing load times.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Basic Computer Components
Subtopic: Physical Storage

Objectives covered
1 Install and Configure Windows
1.1 Recognize basic computer components and explain their functions
1.1.4 Physical storage

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Motherboard

A motherboard is the central printed circuit board (PCB) that connects and allows communication between all computer components. It provides the essential infrastructure for the CPU, memory, storage, expansion cards, and other peripherals to function cohesively. Every motherboard is designed with specific form factors, chipsets, and socket types that determine its compatibility with various hardware components.

Purpose

Upon completing this project, you will better understand the key components of a motherboard.

Steps for Completion

1. A motherboard's core is the CPU socket, which houses the processor and enables communication with other components.
2. Which type of RAM do most modern motherboards support?
    a. DDR4 or DDR5
3. A motherboard contains expansion slots, which allow users to add dedicated graphics cards, sound cards, network adapters, or storage controllers.
4. List at least three types of I/O ports that motherboards provide.
    a. USB, HDMI, and Ethernet
5. A motherboard's chipset determines CPU, RAM, storage, and peripherals communication.
6. ATX motherboards offer the most expandability of all the form factor options, while Mini-ITX boards prioritize compact designs for small form factor (SFF) builds.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Basic Computer Components
Subtopic: Motherboard

Objectives covered
1 Install and Configure Windows
1.1 Recognize basic computer components and explain their functions
1.1.5 Motherboard

Domain 1

Lesson 2

IT Specialist

Device Configuration and Management

Domain User Account

Active Directory (AD) is a crucial directory service used in enterprise environments to manage users, computers, and resources within a network. It allows centralized authentication, authorization, and administration, improving security and efficiency in IT environments. A domain user account is an identity stored in Active Directory that allows users to access network resources. Administrators manage these accounts and provide a secure and scalable way to enforce security policies across an organization.

Purpose

Upon completing this project, you will better understand domain user accounts.

Steps for Completion

1. How do domain user accounts differ from local user accounts?

   a. While local accounts only apply to a user's personal device, a domain account allows a user to access resources seamlessly across devices that are connected to the domain. These are stored in an Active Directory

2. Active Directory organizes user accounts and resources using a hierarchical structure that includes domains, organizational units (OUs), and Group Policies.

3. Through Group Policy Objects (GPOs), administrators can enforce security settings, software installations, and network configurations for users and computers within the domain.

4. Authentication within AD is handled through the Kerberos protocol, ensuring secure and efficient user verification.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Active Directory Services
Subtopic: Domain User Account

Objectives covered
1 Install and Configure Windows
1.2 Describe the purpose of Active Directory services
1.2.1 Domain user account

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Domain Administrator Account

An AD service domain administrator account is a privileged account with full domain control. The account allows IT professionals to manage users, groups, policies, and network resources.

Purpose

Upon completing this project, you will better understand domain administrator accounts, their functions, and best practices for securing them.

Steps for Completion

1. List at least three actions a domain administrator account allows administrators to perform.

   a. Create and manage user accounts
   Configure Group Policy Objects (GPOs)
   Control access permissions
   Manage domain-wide security settings

2. List at least three best practices for securing domain administrator accounts.

   a. Limit usage for everyday tasks
   Implementing multi-factor authentication (MFA)
   Using delegated administration to assign lower-privileged roles
   Regularly auditing account activities

3. List three tools that administrators can use to manage a domain efficiently.

   a. Active Directory Users and Computers (ADUC), PowerShell commands, and Group Policy Management Consol (GPMC)

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Active Directory Services
Subtopic: Domain Administrator Account

Objectives covered
1 Install and Configure Windows
1.2 Describe the purpose of Active Directory services
1.2.2 Domain administrator account

Centralized Management

Active Directory provides IT administrators with a centralized framework for managing user access, computer configurations, and Group Policies across an organization. Active Directory organizes resources hierarchically using domains, organizational units (OUs), and groups.

Purpose

Upon completing this project, you will better understand the centralized management of users, computers, and groups in Active Directory.

Steps for Completion

1. List at least three actions administrators can perform within Active Directory Users and Computers (ADUC).

    a. Create and manage user accounts
    assign permissions
    Enforce security policies

2. Label the following statements as true or false.

    a. True IT teams can enforce software installation policies, apply security updates, and restrict unauthorized access through Group Policy Objects (GPOs).

    b. True Remote Desktop Protocol (RDP) and administrative tools allow IT professionals to troubleshoot issues and manage systems remotely.

    c. False Policies in Active Directory simplify permission management by allowing administrators to assign access rights collectively rather than individually.

    d. False Distribution groups control access to shared resources, while security groups facilitate email communication through Exchange Server.

    e. True Group nesting enhances security by allowing administrators to manage permissions hierarchically.

    f. True Administrators use tools like ADUC, PowerShell, and Group Policy Management Console (GPMC) to streamline user, computer, and group management.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Active Directory Services
Subtopic: Centralized Management

Objectives covered
1 Install and Configure Windows
1.2 Describe the purpose of Active Directory services
1.2.3 Centralized management of users, computers, and groups

Group Policy

Group Policy in AD allows administrators to manage and enforce security settings, configurations, and restrictions across an organization's network. Group Policy enables centralized control over user and computer settings through policies applied at various levels within the AD hierarchy.

Purpose

Upon completing this project, you will better understand Group Policy Objects (GPOs) in AD environments.

Steps for Completion

1. Name some GPO configurations that define system behavior.

   a. Password Policies
   Software installations
   User interface restrictions

2. List at least three actions administrators can perform within the Group Policy Management Console (GPMC).

   a. Restricting USB device access
   Enforcing software updates
   Defining firewall rules

3. Administrators can use tools such as GPMC and PowerShell to manage GPOs efficiently.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Active Directory Services
Subtopic: Group Policy

Objectives covered
1 Install and Configure Windows
1.2 Describe the purpose of Active Directory services
1.2.4 Group Policy

Local and Group Security Policies

Local security policies help enforce rules related to user access, permissions, and system configurations to protect organizational data and resources. Group security policies are managed centrally through Group Policy Objects (GPOs) in Active Directory. These policies are applied to users and computers within a domain, organizational unit (OU), or site. Administrators use tools like the Local Security Policy Editor, Group Policy Management Console (GPMC), and PowerShell to configure and manage security policies.

Purpose

Upon completing this project, you will better understand the differences between local and group security policies.

Steps for Completion

1. Which tool is used to configure local security policies on individual computers?
    a. local Security Policy (secpol.msc)
2. Label the following statements as true or false.
    a. True Local security policies apply to multiple devices.
    b. True Local security policies lack centralized management and scalability.
    c. False Precedence determines which policy is applied when multiple policies exist.
3. List the Group Policy hierarchy used to determine precedence in an AD environment.
    a. Local Policies
    b. Site Policies
    c. Domain Policies
    d. OU Policies
4. Consider a scenario where an organization enforces a password complexity policy. A local security policy requires an 8-character password, while a domain GPO enforces a 12-character password. An administrator configures a password policy in Local Security Policy and Group Policy Management Console. Which password requirement will have the higher precedence?
    a. Group security Policies

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 1
Topic: Active Directory Services
Subtopic: Local and Group Security Policies

Objectives covered
1 Install and Configure Windows
1.2 Describe the purpose of Active Directory services
1.2.5 Differentiate between local and group security policies and precedence

Domain 1

Lesson 3

IT Specialist

Device Configuration and Management

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Installation and Account Options

When setting up Windows 11, key settings include time zone preferences, user account types, and installation methods. These decisions impact usability, security, and system management. Windows 11 offers two primary account types for user authentication: Microsoft accounts and local accounts. Knowing which to use and how to set up these accounts properly benefits both users and administrators.

Purpose

Upon completing this project, you will better understand how to set up Windows 11 accounts and configure settings.

Steps for Completion

1. Navigate to your device's Settings.
2. Set your location and enable automatic time zone detection based on your internet connection.
3. Describe the differences between Microsoft and local accounts.
    a. Microsoft accounts can be synchronized over multiple devices, has cloud integration with OneDrive, the Microsoft store and more security options with password recovery and two-factor authentication. Personal settings can be synced across multiple Windows devices for convenience.

    Local accounts are independent from Microsoft and its services. Basic offline access to Windows 11 is provided which is preferred in privacy-oriented environments or systems that do not need Microsoft services. However, there's no options for password recovery through the Microsoft servers, auto app sync, or cloud-base backup.
4. What kind of user should opt for a Windows 11 upgrade rather than a custom installation?
    a. People who need to preserve their files, apps or settings from Windows 10. This method reduces chances of data loss.
5. What kind of user should opt for a Windows 11 custom installation rather than an upgrade?
    a. People who need a clean slate that has been optimized as per Windows parameters

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 1
Topic: Install Windows
Subtopic: Installation and Account Options

Objectives covered
1 Install and Configure Windows
1.3 Install Windows using the default settings
1.3.1 Time zone options, Microsoft account vs. local account, upgrade vs. custom install

Domain and Non-Domain Devices

Understanding domain-joined and non-domain-joined device installation helps IT professionals configure systems according to business and security requirements.

Purpose

Upon completing this project, you will better understand the differences between domain-joined and non-domain-joined computers.

Steps for Completion

1. Label the following statements as true or false.

    a. True A domain-joined computer is connected to an AD domain, enabling centralized management and security enforcement.

    b. False Non-domain-joined setups are commonly used in enterprise environments.

    c. True A non-domain-joined computer operates independently without AD integration.

    d. False Domain-joined machines are ideal for personal use, small businesses, or workgroups that do not require centralized management.

    e. False Administrators use tools like Active Directory Users and Computers (ADUC), Group Policy Management Console (GPMC), and Windows Settings to configure and manage non-domain-joined computers.

    f. True Non-domain-joined computers may be managed through local policies, Microsoft Intune, or other endpoint management solutions.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Install Windows
Subtopic: Domain and Non-Domain Devices

Objectives covered
1 Install and Configure Windows
1.3 Install Windows using the default settings
1.3.2 Installation scenarios include domain-joined and non-domain-joined computers

Domain 1

Lesson 4

IT Specialist

Device Configuration and Management

LK LearnKey

User Accounts Types

Understanding Microsoft, domain, and local user account types is essential for IT professionals managing user access, security, and authentication in various environments. Microsoft accounts are cloud-based accounts used to access Windows devices and Microsoft services. In enterprise environments, domain accounts are centrally managed through Active Directory (AD). Local accounts exist on a single device and provide access without cloud or domain integration.

Purpose

Upon completing this project, you will better understand Microsoft, domain, and local user account options.

Steps for Completion

1. Number the steps to create a Microsoft account in order.
    a. 3 Confirm the password.
    b. 1 Enter an email address.
    c. 2 Create a password.
2. What do Microsoft accounts enable users to do?
    a. Sync settings across multiple devices, enhance security with multi-factor authentication (MFA), and integrate seamlessly with Microsoft 365 applications
3. What do domain accounts allow users to do?
    a. To authenticate on any domain-joined computer using their credentials while enforcing security policies through Group Policy Objects (GPOs)
4. Why would someone choose to use a local account?
    a. They exist on a single device and be accessed without cloud or domain integration; these are better for stand-alone systems or administrative tasks that require local access control

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: User Account Options
Subtopic: Types of User Accounts

Objectives covered
1 Install and Configure Windows
1.4 Configure user account options
1.4.1 User account (Microsoft, domain, or local)

User and Administrative Accounts

Understanding Standard User and Administrator accounts is vital for configuring devices securely and effectively. Standard User accounts offer limited access and are suitable for everyday tasks, while Administrator accounts provide full control over a system for configuration and management. Best practices for ensuring security and proper system management include limiting administrative access and enforcing strong password policies.

Purpose

Upon completing this project, you will better understand the differences between Standard User and Administrator accounts.

Steps for Completion

1. Navigate to your device's Settings.
2. If you have the permission to do so, go through the steps to create a Standard User account on your device. You do not have to finish the process.
3. If you have the permission to do so, go through the steps to create an Administrator account on your device. You do not have to finish the process.
4. Administrator accounts can modify system settings, install software, and manage other users, making them responsible for device configuration and maintenance.
5. Standard user accounts have limited access to system resources. They can use applications, modify personal settings, and perform other tasks within their assigned permissions, but cannot install software or change system-wide settings.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: User Account Options
Subtopic: User and Administrative Accounts

Objectives covered
1 Install and Configure Windows
1.4 Configure user account options
1.4.2 Standard user and administrative account types

LK LearnKey

User Profiles

User profiles consist of settings and preferences associated with individual user accounts. They are critical in providing personalized and secure computing experiences. Whether using local profiles for individual devices or roaming profiles for more extensive environments, understanding how to manage user profiles is important for personalizing user experiences while maintaining system security and consistency.

Purpose

Upon completing this project, you will better understand user profiles, essential device configuration, and management.

Steps for Completion

1. List at least three settings that can be personalized within a user account.

    a. Desktop settings
    Preferences for applications
    System configurations specific to that user

2. A local profile is stored on a specific device and contains the user's settings and files for that particular system. This type of profile is ideal for users who only use one device.

3. A roaming profile is typically used in an enterprise environment. This type of profile is stored on a network server, allowing users to sign in to any device on the network and access their personalized settings and files.

4. When setting up user profiles, which factors should be considered?

    a. The environment (personal vs. enterprise), security requirements, and data synchronization across devices
    Local accounts are better for personal use, but in larger environments, especially if the user is moving between devices and locations, roaming is better for flexibility and consistency.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: User Account Options
Subtopic: User Profiles

Objectives covered
1 Install and Configure Windows
1.4 Configure user account options
1.4.3 User profiles

Domain 1

Lesson 5

IT Specialist

Device Configuration and Management

Start Menu

Configuring desktop settings, including the Start menu, is essential to device management. Customizing the Start menu layout, pinning frequently used items, and adjusting taskbar and notification area settings can enhance user experience and productivity. Understanding the various options for configuring these settings ensures that users can work more efficiently and comfortably with a desktop environment tailored to their needs.

Purpose

Upon completing this project, you will better understand how to configure your Start menu and overall workspace.

Steps for Completion

1. Navigate to your device's Settings.
2. Personalize the behavior and appearance of your Start menu.
    a. Customize the tile layout.
    b. Choose to show your recently-added and most-used apps.
3. Pin a shortcut to an application you use frequently to the Start menu.
4. Customize your taskbar by pinning and unpinning apps you use frequently and less often, respectively.
5. Adjust your notification settings.
6. virtual desktops allow users to organize their workspace by creating multiple desktop environments for tasks or projects.

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 1
Topic: Desktop Settings
Subtopic: Start Menu

Objectives covered
1 Install and Configure Windows
1.5 Configure desktop settings
1.5.1 Start menu

Display Settings

Configuring display settings ensures users can interact comfortably and efficiently with their devices. Display settings control how content is shown on a screen, including resolution, scaling, orientation, and multiple display configurations. Proper display configuration is crucial for maintaining visual clarity, user comfort, and device performance.

Purpose

Upon completing this project, you will better understand how to configure display settings.

Steps for Completion

1. Navigate to your device's Settings.
2. What is your device's display resolution setting?
    a. 1920 x 1200
3. Adjust the scaling of your screen to your preference.
4. Set the orientation of your screen to your preference.
5. Set the Night Light setting to your preference.
6. Review the advanced display settings and set them to your preference.
7. Label the following statements as true or false.
    a. False Too high a resolution can result in blurry or stretched images, while too low a resolution may cause elements on a screen to appear too small.
    b. True Landscape mode is the most common orientation, but portrait mode can be helpful for tasks like reading long documents or working with code.
    c. True Most operating systems allow users to connect two or more monitors and extend or duplicate a display across them.
    d. True Users can independently adjust each monitor's resolution and orientation in a multi-display setup.
    e. False The Night Light setting does not reduce eye strain.
    f. True A higher refresh rate can improve the smoothness of motion, while using more energy.
    g. True Color calibration settings adjust a display's color accuracy for more precise visual work.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 1
Topic: Desktop Settings
Subtopic: Display Settings

Objectives covered
1 Install and Configure Windows
1.5 Configure desktop settings
1.5.2 Display settings

Application Shortcuts

Application shortcuts are small icons or links that point to an application, file, or folder, providing quick access. They are typically found on the desktop, taskbar, or within the Start menu. Users can also create shortcuts to files and folders that they use often. Users can significantly enhance their productivity by customizing and organizing their shortcuts. Keyboard shortcuts, also known as predefined key combinations, can also enhance user productivity.

Purpose

Upon completing this project, you will better understand application shortcuts.

Steps for Completion

1. Create a desktop shortcut to an application you use often.
2. Pin an application you use often to the taskbar.
3. Pin an application you use often to the Start menu.
4. Create a desktop shortcut to a folder or files you access often.
5. In Windows, what does the keyboard shortcut Ctrl + Shift + Esc open?
    a. Task Manager
6. Create a custom keyboard shortcut.
7. What is your shortcut, and what does it do?
    a. ctrl + alt + o
8. Delete any shortcuts you do not want on the desktop.
9. Unpin any applications you do not want on the taskbar and/or Start menu.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 1
Topic: Desktop Settings
Subtopic: Application Shortcuts

Objectives covered
1 Install and Configure Windows
1.5 Configure desktop settings
1.5.3 Application shortcuts

Time Zone Settings

Time Zone settings ensure that devices display accurate time across different regions and maintain synchronization with global standards. Time zone settings are critical for keeping devices in sync with local time, managing schedules, and enabling time-dependent features like alarms, automatic updates, and system logs. Devices must be configured to reflect the correct time zone to avoid issues like incorrect time stamps on emails, documents, or system logs.

Purpose

Upon completing this project, you will better understand how to configure time zone settings.

Steps for Completion

1. Navigate to your device's Settings.
2. Review Date & time settings and select your preferred time zone.
3. What time zone are you in?
    a. (UTC 06:00) Central Time (US & Canada)
4. If desired, set the automatic time zone feature.
5. Select any other time settings you prefer.
6. What is the importance of the time synchronization feature?
    a. It is crucial for keeping accurate times for processes such as system logs, scheduling tasks, and coordinating with other systems
7. List at least two applications that rely on accurate time to schedule events or meetings.
    a. Calendar applications
    email cliants
    messaging apps

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 1
Topic: Desktop Settings
Subtopic: Time Zone Settings

Objectives covered
1 Install and Configure Windows
1.5 Configure desktop settings
1.5.4 Time zone settings

Domain 1

Lesson 6

IT Specialist

Device Configuration and Management

LK LearnKey

Taskbar Personalization

The taskbar is a central feature of most operating systems, providing quick access to applications, system notifications, and important features such as the Start menu, Search bar, and clock. Customizing the taskbar according to user preferences can help streamline workflows and improve productivity.

Purpose

Upon completing this project, you will better understand how to customize various aspects of a taskbar.

Steps for Completion

1. Navigate to your device's Settings.
2. Customize your taskbar.
    a. Set the taskbar's alignment to your preference.
    b. Select your preferred taskbar behaviors.
    c. Set your preferred Search display.
    d. Customize your notification icon display as desired.
    e. Pin a frequently used application to the taskbar.
    f. Unpin any unwanted applications on the taskbar.
    g. If desired, set your Windows surfaces to appear translucent or change the accent color.
    h. Adjust your taskbar size to your preference.
3. Take a screenshot of your personalized taskbar.
4. Save the screenshot as **155-Completed.png** to your Domain 1 Student folder.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 1
Topic: Taskbar and Power Settings
Subtopic: Taskbar Settings; Taskbar Personalization

Objectives covered
1 Install and Configure Windows
1.5 Configure desktop settings
1.5.5 Taskbar settings

Power Settings

Power settings in Windows 11 play a crucial role in managing energy consumption, optimizing performance, and extending battery life. Users can balance performance and energy efficiency by properly configuring power settings and features.

Purpose

Upon completing this project, you will better understand a device's power settings.

Steps for Completion

1. Navigate to your device's Settings.
2. Which power plan is your device set to use?
   a. Balanced
3. Match the power setting or feature to its use.

A. Screen timeout setting	C. Hibernate mode
B. Energy saver mode	D. Sleep mode

   a. D Helps reduce power consumption by putting a device into a low-power state after a period of inactivity
   b. A Controls how long a display remains on before turning off, helping conserve energy when idle
   c. C Helps preserve a system state while using minimal power, as it saves the current session to disk and powers the device off completely
   d. B Extends battery life by limiting background processes and reducing screen brightness

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Taskbar and Power Settings
Subtopic: Power Settings

Objectives covered
1 Install and Configure Windows
1.5 Configure desktop settings
1.5.6 Power settings

Window Management

Window management helps users efficiently organize and interact with multiple applications. Minimizing, closing, and snapping windows improves workflow and enables seamless multitasking.

Purpose

Upon completing this project, you will better understand how to manage windows.

Steps for Completion

1. Open a search engine.
2. Minimize the search engine window.
3. Maximize the search engine window.
4. Snap the search engine window to one half of the screen.
5. Close the search engine window.
6. Minimizing a window hides it from view without closing it, freeing up screen space while keeping the application accessible from the taskbar.
7. Closing a window exits the program and frees system resources, though some applications may continue running in the background.
8. Snapping windows allows users to quickly arrange multiple applications by dragging them to screen edges, automatically resizing them for side-by-side viewing.
9. Virtual desktops allow users to create multiple desktops with different open applications.

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 1
Topic: Taskbar and Power Settings
Subtopic: Windows Management

Objectives covered
1 Install and Configure Windows
1.5 Configure desktop settings
1.5.7 Window management (minimize, close, snap)

Domain 1

Lesson 7

IT Specialist

Device Configuration and Management

LK LearnKey

Mouse Settings

Mouse settings are an essential aspect of accessibility that help users customize their mouse for a comfortable and efficient experience. These settings benefit individuals with disabilities or specific preferences, ensuring better usability and control. Some mouse settings are found in a device's Settings; others can be adjusted within the Control Panel.

Purpose

Upon completing this project, you will better understand mouse settings.

Steps for Completion

1. Navigate to your device's Settings.
2. Set your preferred mouse pointer
    a. You may choose to change the pointer size and color.
3. Locate the Mouse keys option and try it out.
    a. Adjust the mouse keys' speed and acceleration settings.
    b. You can toggle off this setting if you prefer not to use it.
4. Adjust your mouse pointer's speed to your liking.
5. Who might benefit from swapping the primary and secondary mouse buttons?
    a. People with limited mobility or are left-handed
6. Open your device's Control Panel.
7. Locate and review the Double-click speed and ClickLock settings.
8. Customize your mouse's scrolling behavior to fit your preference.
9. Adjust your mouse's pointer speed using the Enhance pointer precision option if desired.
10. Who might benefit from utilizing the Double-click speed setting?
    a. People with tremors or limited dexterity
11. Who might benefit from utilizing the ClickLock feature?
    a. People with difficulties holding down buttons for extended periods of time

Project Details

Project file
N/A

Estimated completion time
10-15 minutes

Video reference
Domain 1
Topic: Accessibility Settings
Subtopic: Mouse Settings; Mouse Settings in the Control Panel

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.1 Mouse Settings

Color Filters and Contrast Themes

Color filters and contrast themes help users with visual impairments by altering on-screen colors to improve visibility and contrast. Windows 11 offers built-in color filters and contrast themes that adjust a screen's color profile to create a more accessible experience. Understanding how and when to enable and customize these settings greatly aids inclusivity and improves usability.

Purpose

Upon completing this project, you will better understand color filters and contrast themes.

Steps for Completion

1. Navigate to your device's Settings.
2. Preview and apply different color filters. Select your favorite.
3. List at least three types of color filters.
    a. greyscale, inverted colors, and specialized colorblindness filters
4. Preview and apply different contrast themes. Select your favorite.
5. List at least three types of contrast themes.
    a. Aquatic, desert, night sky

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Accessibility Settings
Subtopic: Color Filters; Contrast Themes

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.2 Color filters
1.6.3 Contrast themes

Audio Settings

Audio settings are a crucial part of system configuration in Windows 11 that enhances sound quality, accessibility, and usability. Properly configuring audio settings ensures an optimal sound experience for communication, media playback, and accessibility needs. Understanding how to configure and troubleshoot audio settings ensures people experience clear, high-quality sound across various applications.

Purpose

Upon completing this project, you will better understand how to configure audio settings in Windows.

Steps for Completion

1. In Windows Sound settings, turn on Mono audio.
2. Take a screenshot with this setting enabled.
3. Save the file as **164-Completed.png** to your Domain 1 Student folder.
4. What tool can be used if there are issues with sound?
    a. Troubleshoot
5. Is a microphone an Output device or an Input device?
    a. Input device

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Accessibility Settings
Subtopic: Audio Settings

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.4 Audio Settings

Magnifier

Magnifiers are great for seeing things more clearly, and, in Windows 11, the Magnifier is an essential accessibility tool designed to help users with visual impairments by enlarging screen content for better readability. Magnifier enhances usability by providing different zoom levels, viewing modes, and customization options. Proper configuration ensures an accessible and comfortable viewing experience.

Purpose

Upon completing this project, you will better understand how to use the magnifier in Windows.

Steps for Completion

1. Navigate to Magnifier in Windows Accessibility settings.
2. Change the Zoom level to 1500%.
3. Take a screenshot showing the new Magnifier setting.
4. Save the screenshot as **165-Completed.png** to your Domain 1 Student folder.
5. Which Magnifier mode creates a movable magnified area?
    a. Lens mode
6. What is the keyboard shortcut to activate Magnifier?
    a. Windows + Plus (+)

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Accessibility Settings
Subtopic: Magnifier

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.5 Magnifier

Narrator

Narrator is a built-in screen reader in Windows 11 designed to assist users with visual impairments by reading aloud on-screen text and providing navigation guidance. Narrator enables users to interact with their computer without relying on visual elements, making digital content more accessible. It reads headings, links, and text aloud, helping users interact with the content more effectively.

Purpose

Upon completing this project, you will better understand how to use the Narrator to enhance accessibility.

Steps for Completion

1. Turn on Narrator in Windows Accessibility settings.
2. Take a screenshot showing Narrator turned on.
3. Save the screenshot as **166-Completed.png** to your Domain 1 Student folder.
4. What is the keyboard shortcut to turn on Narrator?
    a. Ctrl + Windows + Enter

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Accessibility Settings
Subtopic: Narrator

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.6 Narrator

Domain 1

Lesson 8

IT Specialist

Device Configuration and Management

Sticky Keys

Sticky keys are an accessibility feature built into most operating systems, including Windows 11. This feature helps users with keyboard shortcuts by allowing them to select keys sequentially rather than simultaneously. Sticky keys are particularly useful for users with mobility impairments or those using adaptive input devices.

Purpose

Upon completing this project, you will become more familiar with sticky keys.

Steps for Completion

1. Enable sticky keys in Windows Accessibility settings.
2. Take a screenshot showing that sticky keys are enabled.
3. Save the file as **167-Completed.png** to your Domain 1 Student folder.
4. Which key on the keyboard can users select five times to enable sticky keys?
    a. _____
5. What are two additional customization options for sticky keys in Windows 11?
    a. _____
    _____
    _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Keyboard, Voice, and Cursor Settings
Subtopic: Sticky Keys

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.7 Sticky Keys

On-Screen Keyboard

Another accessibility tool available in Windows 11 is the on-screen keyboard (OSK). This tool provides a virtual keyboard on the screen and is designed for users who have difficulty using a physical keyboard. OSKs ensure users can enter text, navigate applications, and perform other essential computing tasks without a traditional keyboard.

Purpose

Upon completing this project, you will better understand on-screen keyboards.

Steps for Completion

1. Why might someone need to use an on-screen keyboard?
    a. _____
2. Enable the on-screen keyboard in Windows 11 Accessibility settings.
3. Take a screenshot showing that the OSK is enabled.
4. Save the file as **168-Completed.png** to your Domain 1 Student folder.
5. What is the keyboard shortcut for enabling the OSK in Windows 11?
    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Keyboard, Voice, and Cursor Settings
Subtopic: On-Screen Keyboard

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.8 On-screen keyboard

Voice Access and Voice Typing

A robust accessibility feature in Windows 11 is voice access, which allows users to control their PC and dictate text using only their voice. This feature assists individuals with mobility impairments, those recovering from injuries, or users who favor hands-free interaction for greater efficiency and convenience.

Voice typing is a built-in speech-to-text feature in Windows 11 that allows users to dictate text rather than typing it manually. This feature allows users to convert spoken words into text in any application that contains a text field.

Purpose

Upon completing this project, you will become more familiar with voice access and voice typing in Windows 11.

Steps for Completion

1. Enable voice access in the Windows 11 Accessibility settings.
2. Take a screenshot showing that this feature is enabled.
3. Save the file as **169-Completed.png** to your Domain 1 Student folder.
4. Enable voice typing in the Windows 11 Accessibility settings.
5. Take a screenshot showing that this feature is enabled.
6. Save the file as **1610-Completed**
7. What is the keyboard shortcut for activating voice typing in Windows 11?
    a. _____

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 1
Topic: Keyboard, Voice, and Cursor Settings
Subtopic: Voice Access; Voice Typing

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.9 Voice access
1.6.10 Voice typing (dictation)

Text Cursor and Eye Control

The text cursor in Windows 11 can be configured to enhance usability and accessibility. The text cursor is the blinking indicator showing where text will appear when typing. Several customization options are available in Windows 11 to improve visibility, making it easier for users to locate and track the cursor while typing.

Another accessibility feature in Windows 11 is Eye Control, which allows users to navigate their computers using only eye movements. This technology benefits individuals with limited mobility who cannot use traditional input devices. Eye Control enables users to move the cursor, select items, and type text with their eyes.

Purpose

Upon completing this project, you will better understand the text cursor and Eye Control settings in Windows 11.

Steps for Completion

1. In the Windows 11 Accessibility settings, change the text cursor size to 5 and the thickness to 6.
2. Change the text cursor color to one of your choosing.
3. Take a screenshot of the settings you have changed.
4. Save the file as **1611-Completed.png** to your Domain 1 Student folder.
5. Navigate to Eye Control.
6. Take a screenshot showing the Eye Control screen.
7. Save the file as **1612-Completed**
8. What does Eye Control require to be fully functional on a computer?
    a. _____
9. What are two key features of the Eye Control interface?
    a. _____
    _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 1
Topic: Keyboard, Voice, and Cursor Settings
Subtopic: Text Cursor; Eye Control

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.11 Text cursor
1.6.12 Eye Control

Live Captions

Live Captions are a built-in accessibility feature in Windows 11 that provides real-time audio and video content captions. This feature supports individuals who are deaf or hard of hearing and users in noisy environments.

Purpose

Upon completing this project, you will better understand Live Captions in Windows 11.

Steps for Completion

1. In the Windows 11 Accessibility settings, activate Live Captions.
2. Take a screenshot showing Live Captions enabled.
3. Save the file as **1613-Completed.png** to your Domain 1 Student folder.
4. What is the Windows keyboard shortcut for enabling Live Captions?
    a. _____
5. Label the following statements as true or false.
    a. _____ Live Captions only work on video players like YouTube, Netflix, and VLC Media Player.
    b. _____ Users can adjust the position, size, font style, and background of Live Captions.
    c. _____ Windows 11 Live Captions only support the English language.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: Keyboard, Voice, and Cursor Settings
Subtopic: Live Captions

Objectives covered
1 Install and Configure Windows
1.6 Configure accessibility settings to meet users' specific needs
1.6.13 Live Captions

Domain 1

Lesson 9

IT Specialist

Device Configuration and Management

Windows Update Settings

Windows 11 Update ensures a system receives the latest patches, security fixes, and feature improvements. Keeping Windows 11 up to date is important for security, performance, and stability. Windows can automatically check for updates, or users can do it manually.

Purpose

Upon completing this project, you will become more familiar with Windows Update settings.

Steps for Completion

1. Open Windows Update and manually check for updates. If any are available, download and install them.
2. Take a screenshot showing Windows is up to date.
3. Save the file as **171a-Completed.png** to your Domain 1 Student folder.
4. Manually adjust the active hours for restarting after an update.
5. Adjust to a start time of 7:00 AM and an end time of 4:00 PM
6. Take a screenshot of the adjusted hours.
7. Save the file as **171b-Completed**
8. Label the following statements as true or false.
    a. _____ It is risky to delay updates for Windows.
    b. _____ If an update causes issues, users can uninstall and restore their system to a previous state.

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 1
Topic: OS and App Updates
Subtopic: Windows Update Settings

Objectives covered
1 Install and Configure Windows
1.7 Manage updates
1.7.1 Windows Update Settings

LK LearnKey

Software Updates and Patches

Software updates fix bugs, improve performance, and add new features, while security patches address specific vulnerabilities that could be exploited by malware or hackers. Users must update their software and systems to protect them against the latest threats. Staying up to date also ensures a system benefits from ongoing improvements.

Purpose

Upon completing this project, you will better understand software updates and patches.

Steps for Completion

1. Check for software updates in Windows 11.
    a. If any are available, download and install them.
2. Take a screenshot showing that no new updates are available.
3. Save the file as **172a-Completed.png** to your Domain 1 Student folder.
4. Check for third-party application updates in the Microsoft Store.
    a. If any are available, download and install them.
5. Take a screenshot showing there are no new third-party application updates.
6. Save the file as **172b-Completed**
7. Label the following statements as true or false.
    a. _____ Users must manually apply security patches when updating Windows 11.
    b. _____ Users can delay updates for up to 35 days.

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 1
Topic: OS and App Updates
Subtopic: Software Updates and Patches

Objectives covered
1 Install and Configure Windows
1.7 Manage updates
1.7.2 Software updates and patches

Optional OS Updates

Some OS updates are optional, meaning they are not essential for system security, but they can enhance hardware performance and address minor software issues. Regularly reviewing and applying relevant optional updates helps keep a system running smoothly.

Purpose

Upon completing this project, you will become more familiar with optional OS updates.

Steps for Completion

1. Search for optional updates in Windows 11.
2. Take a screenshot of the result.
3. Save the file as **173-Completed.png** to your Domain 1 Student folder.
4. What is included in an optional update?
    a. _____
    _____
5. How might optional updates be useful for a printer that is not responding?
    a. _____
6. Label the following statements as true or false.
    a. _____ Driver updates are mandatory.
    b. _____ An optional update can be removed by reverting to a previous state.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: OS and App Updates
Subtopic: Optional OS Updates

Objectives covered
1 Install and Configure Windows
1.7 Manage updates
1.7.3 Optional OS updates

Device Driver Updates

Keeping device drivers updated ensures that hardware will function correctly, improves performance, and prevents compatibility issues. Regularly checking for driver updates and knowing when to install manufacturer drivers ensures that a system remains optimized for its best performance and reliability.

Purpose

Upon completing this project, you will better understand device driver updates.

Steps for Completion

1. What is the function of device drivers?
    a. _____
2. What are the two ways to check for driver updates?
    a. _____
3. What do manufacturer-provided drivers often include?
    a. _____
4. If a manufacturer driver causes issues, what can one do and how?
    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: OS and App Updates
Subtopic: Device Driver Updates

Objectives covered
1 Install and Configure Windows
1.7 Manage updates
1.7.4 Device driver updates (Microsoft and manufacturer)

Update History

All installed updates are recorded in Windows 11 through the Update History feature. This feature provides a detailed log of system updates, allowing users to review past updates, troubleshoot issues, and uninstall problematic updates if necessary. Regularly viewing the Update History keeps users informed of system changes and helps them manage system stability effectively.

Purpose

Upon completing this project, you will become more familiar with Update History.

Steps for Completion

1. What are examples of the system updates included in Update History's detailed log?

    a. _____

2. Open the Update History in Windows Update.
3. Take a screenshot of the information shown there.
4. Save the file as **175-Completed.png** to your Domain 1 Student folder.
5. Where can one uninstall a problematic update in Update History?

    a. _____

6. What tool can be used to detect and fix update-related issues automatically?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 1
Topic: OS and App Updates
Subtopic: Update History

Objectives covered
1 Install and Configure Windows
1.7 Manage updates
1.7.5 Update history

Domain 2

Lesson 1

IT Specialist

Device Configuration and Management

Install Applications

Installing applications in Windows 11 requires understanding user account permissions and security settings to ensure proper software management. Understanding user account permissions and security settings is essential for maintaining control over software installations in Windows 11. User Account Control (UAC) prompts, Microsoft Store applications, and Group Policy settings help ensure that only authorized users install applications while keeping a system secure and functional.

Purpose

Upon completing this project, you will better understand how to install applications.

Steps for Completion

1. What is the difference between Administrator accounts and Standard User accounts?

    a. _____

2. What applications on the Microsoft Store may require administrative approval?

    a. _____

3. What feature can administrators use to restrict application installations?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Manage Apps
Subtopic: User Account Requirements

Objectives covered
2 Features, Apps, and Peripherals
2.1 Manage applications and Windows features
2.1.1 Identify user account requirements and permissions for application installation

Modify Applications

Windows 11 provides multiple ways to modify applications, whether through the Settings application, Control Panel, or the application's settings. Modifying applications allows users to update settings, add or remove features, repair issues, or uninstall them.

Purpose

Upon completing this project, you will better understand how to modify applications.

Steps for Completion

1. Label the following statements as true or false.

    a. _____ There are no applications that allow users to remove specific components without reinstalling the programs.

    b. _____ Modification options are available for some desktop applications through the Control Panel.

    c. _____ If an application is not working correctly, repairing it is an alternative to uninstalling.

    d. _____ Legacy applications can be uninstalled through the Apps page.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Manage Apps
Subtopic: Modify App Installations

Objectives covered
2 Features, Apps, and Peripherals
2.1 Manage applications and Windows features
2.1.2 Modify application installations

Remove Desktop Applications

Windows provides multiple methods to remove applications, including through Settings, Control Panel, and the Start Menu, ensuring users have flexibility in managing their installed programs.

Purpose

Upon completing this project, you will better understand how to remove desktop applications.

Steps for Completion

1. What are three benefits of uninstalling desktop applications?

    a. _____
    _____
    _____

2. Label the following statements as true or false.

    a. _____ Applications installed via the Microsoft Store can be removed directly from the Start menu.

    b. _____ There are built-in tools that can ensure that all leftover files and registry entries are deleted.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Manage Apps
Subtopic: Remove Desktop Apps

Objectives covered
2 Features, Apps, and Peripherals
2.1 Manage applications and Windows features
2.1.3 Remove desktop application

Domain 2

Lesson 2

IT Specialist

Device Configuration and Management

Optional Features

Windows 11 includes optional features that can be turned on or off based on user needs. These features include system utilities such as Windows Subsystem for Linux (WSL), Hyper-V, and Internet Explorer Mode. These are not always included by default, but can be manually enabled when required.

Purpose

Upon completing this project, you will better understand the optional features available in Windows 11.

Steps for Completion

1. Label the following statements as true or false.

    a. _____ Advanced system features can be found in the Control Panel.

    b. _____ All features require a system restart to complete installation.

2. Match the optional Windows feature to its description.

A. Windows Subsystem for Linux (WSL)	C. Telnet Client
B. Hyper-V	

    a. _____ A legacy tool used for remote communication

    b. _____ Provides virtualization capabilities

    c. _____ Allows users to run Linux distributions alongside Windows

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Manage Windows Features
Subtopic: Optional Windows Features

Objectives covered
2 Features, Apps, and Peripherals
2.1 Manage applications and Windows features
2.1.4 Locate and identify optional Windows features

The Microsoft Store

The Microsoft Store is the official digital distribution platform for Windows applications, offering a secure and convenient way to download and manage apps, games, and digital content. It provides a centralized platform where users can find verified software, ensuring safety and compatibility with Windows 11.

Purpose

Upon completing this project, you will better understand the purpose of the Microsoft Store.

Steps for Completion

1. Label the following statements as true or false.
    a. _____ Users can browse applications by category.
    b. _____ Microsoft vets all applications in the Store for security and performance.
    c. _____ Applications from the Microsoft Store must be updated manually.
    d. _____ The Microsoft Store requires bank account details to make purchases.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Manage Windows Features
Subtopic: Microsoft Store Purpose

Objectives covered
2 Features, Apps, and Peripherals
2.1 Manage applications and Windows features
2.1.5 Describe the purpose of the Microsoft Store

Application Storage

Windows 11 organizes installed applications based on system architecture, distinguishing between 32-bit (x86) and 64-bit (x64) applications. Understanding where applications are stored helps with troubleshooting, system organization, and efficient storage management.

Purpose

Upon completing this project, you will better understand how applications are stored.

Steps for Completion

1. In what folder are 32-bit applications stored?
    a. _____
2. In what folder are 64-bit applications stored?
    a. _____
3. Where are all applications stored on a 32-bit system?
    a. _____
4. Why is the location of Microsoft Store applications unable to be accessed by users?
    a. _____
    _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Manage Windows Features
Subtopic: App Default Locations

Objectives covered
2 Features, Apps, and Peripherals
2.1 Manage applications and Windows features
2.1.6 Understand default locations for applications based on architecture

Remote Desktop

Windows 11 includes Remote Desktop, a feature that allows users to access and control another computer remotely. This functionality is useful for IT support, remote work, and troubleshooting, enabling users to connect to a distant computer as if they were physically present at the device.

Purpose

Upon completing this project, you will better understand Remote Desktop.

Steps for Completion

1. What Windows feature may prevent remote connections if a certain setting is not enabled?

    a. _____

2. Label the following statements as true or false.

    a. _____ If prompted, users should enter the username and password of the remote machine, ensuring they use credentials with administrator access when creating a remote connection.

    b. _____ Users cannot access a Windows machine from macOS, Linux, or a mobile device.

    c. _____ Lowering visual effects in the Experience tab improves performance on slow connections.

    d. _____ Users should ensure no VPNs (Virtual Private Networks) are enabled when making remote connections.

    e. _____ To end a session, users can either disconnect by closing the Remote Desktop window or sign out of the remote session.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Manage Windows Features
Subtopic: Remote Desktop

Objectives covered
2 Features, Apps, and Peripherals
2.1 Manage applications and Windows features
2.1.7 Remote Desktop

Domain 2

Lesson 3

IT Specialist

Device Configuration and Management

Using Copilot

Windows 11 includes various built-in troubleshooting tools to help users quickly diagnose and resolve system issues. Windows Copilot can launch troubleshooting utilities, suggest solutions, and guide users through resolving common problems. Copilot enhances user productivity by providing a conversational interface for opening applications, adjusting settings, generating content, translating text, writing code, and retrieving information. Copilot is pre-installed in Windows 11, making it readily available for all users without requiring separate downloads or configurations.

Purpose

Upon completing this project, you will better understand how to use Windows Copilot.

Steps for Completion

1. Label the following statements as true or false.
    a. _____ Copilot can direct users to the Task Manager to update or reinstall drivers to fix hardware-related issues.
    b. _____ Copilot integrates with Microsoft Edge.
    c. _____ Copilot can access personal files without permission from users.
    d. _____ Copilot can open applications such as Microsoft Word.
    e. _____ Users can summarize text with Copilot.
    f. _____ Copilot can translate text into multiple languages.
    g. _____ Copilot cannot be used to write programming code.
    h. _____ Copilot receives regular updates via Windows Update.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: Windows Copilot
Subtopic: Troubleshooting Directions

Objectives covered
2 Features, Apps, and Peripherals
2.2 Explain the purpose and capabilities of Windows Copilot
2.2.1 Direct users to troubleshooting tools within Windows 11

Copilot Limitations

Windows Copilot is a valuable AI-powered assistant that enhances productivity and system management. However, users should know that it is not a complete replacement for traditional troubleshooting tools, IT support, or advanced automation solutions. By combining Copilot's capabilities with Windows' built-in utilities and professional IT practices, users can achieve a more comprehensive and practical approach to system optimization and problem resolution.

Purpose

Upon completing this project, you will better understand the limitations of Windows Copilot.

Steps for Completion

1. What is required for Copilot to access its full range of AI-driven capabilities?

   a. _____

2. How is Copilot limited in its troubleshooting capabilities?

   a. _____
   _____
   _____

3. How is Copilot limited in its ability to make big system changes for enterprise-level troubleshooting?

   a. _____
   _____
   _____

4. What are three ways Copilot's content may be inaccurate?

   a. _____
   _____

5. What are two alternatives to Copilot for advanced automation?

   a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: Windows Copilot
Subtopic: Uses for Copilot; Default Installation; Limitations of Copilot

Objectives covered
2 Features, Apps, and Peripherals
2.2 Explain the purpose and capabilities of Windows Copilot
2.2.2 Can open applications, access Settings, generate content, translate text into different languages, generate programming code and retrieve information
2.2.3 Installed with Windows 11 by default
2.2.4 Limitations of Copilot

Domain 2

Lesson 4

IT Specialist

Device Configuration and Management

LK LearnKey

The Windows Registry

The Windows Registry is a centralized database that stores and manages critical information related to user preferences, application settings, and Windows system configurations. It ensures that Windows and installed applications operate smoothly by maintaining structured data that defines how software and hardware function.

Purpose

Upon completing this project, you will better understand the purpose of the Windows Registry.

Steps for Completion

1. Match each Windows Registry root key to its description.

A. HKEY_CURRENT_USER	B. HKEY_LOCAL_MACHINE

   a. _____ Contains system-wide settings, including driver configurations, startup programs, and installed software details

   b. _____ Stores individual user settings, such as themes, screen resolutions, and accessibility options

2. What can happen if incorrect changes are applied to the Windows Registry?

   a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Windows Registry
Subtopic: Preferences and Settings

Objectives covered
2 Features, Apps, and Peripherals
2.3 Describe the purpose of the Windows Registry
2.3.1 Database of user preferences, application settings, Windows Settings

HDMI

High-Definition Multimedia Interface (HDMI) is the most widely used digital video and audio connection standard, transmitting high-definition video and multi-channel audio over a single cable. HDMI comes in three sizes: Full HDMI (Type A), Mini HDMI (Type C), and Micro HDMI (Type D), each designated for different devices.

Purpose

Upon completing this project, you will better understand the different types of HDMI connections.

Steps for Completion

1. Match the HDMI connection to its description.

A. Full HDMI (Type A)	C. Micro HDMI (Type D)
B. Mini HDMI (Type C)	

   a. _____ A smaller variant designed for DSLR cameras, tablets, and compact laptops

   b. _____ A compact variant used in action cameras, small digital cameras, and select tablets

   c. _____ The standard connector on TVs, monitors, gaming consoles, streaming devices, and home theater systems

2. List one limitation of HDMI connectors.

   a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Video Connection Types
Subtopic: HDMI

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.1 HDMI (full, mini and micro)

LK LearnKey

DisplayPort

DisplayPort is a high-performance digital display interface designed for superior video and audio transmission, primarily in professional and high-performance computing. Unlike HDMI, which is geared toward consumer electronics, DisplayPort excels in PC setups with higher resolutions, refresh rates, and multi-monitor support.

Purpose

Upon completing this project, you will better understand DisplayPort.

Steps for Completion

1. What is multi-stream transport (MST)?
    a. _____
2. What two adaptive sync technologies eliminate screen tearing?
    a. _____
3. What resolution and refresh rate does DisplayPort 1.2 support?
    a. _____
4. What resolution and refresh rate does DisplayPort 1.4 support?
    a. _____
5. What resolution and refresh rate does DisplayPort 2.0 support?
    a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: Video Connection Types
Subtopic: DisplayPort

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.2 DisplayPort (full, mini)

DVI

The Digital Visual Interface (DVI) is a widely used video connection standard that bridges the gap between analog and digital display technologies. It was developed to provide better image quality than VGA while maintaining compatibility with digital and analog signals. There are three main types of DVI connectors: DVI-D (Digital Only), DVI-A (Analog Only), and DVI-I (Integrated Digital and Analog).

Purpose

Upon completing this project, you will better understand the purpose of Digital Visual Interface (DVI) connectors.

Steps for Completion

1. Match the DVI connector to its description.

A. DVI-D	B. DVI-A	C. DVI-I

   a. _____ Supports analog signals and mainly connects older CRT monitors that require analog input

   b. _____ Combines digital and analog capabilities, making it versatile for users who need a connection supporting either signal type

   c. _____ Designed for digital displays, transmitting pure digital signals without conversion. This connector produces sharper, clearer images, making it ideal for LCD monitors and high-definition applications

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Video Connection Types
Subtopic: DVI

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.3 DVI family of connectors

Domain 2

Lesson 5

IT Specialist

Device Configuration and Management

VGA

Video Graphics Array (VGA) is one of the oldest monitor connection standards, initially introduced in 1987. VGA transmits an analog video signal, which is lower quality than a digital video signal. VGA does not support as high a resolution as digital video connections.

Purpose

Upon completing this project, you will better understand the purpose of Video Graphics Array (VGA) connectors.

Steps for Completion

1. What issues can occur with VGA connectors?
    a. _____
    _____
2. What three types of items use VGA connectors?
    a. _____
    _____
3. Label the following statement as true or false.
    a. _____ VGA transmits audio.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: VGA, USB, and Thunderbolt
Subtopic: VGA

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.4 VGA

USB-A, Mini-A, Micro-A, and USB 3.0

USB-A, Mini-A, and Micro-A are variations of the Universal Serial Bus (USB) standard, designed for data transfer and power delivery. USB has evolved to offer faster speeds, better connectivity, and greater versatility. USB-A is the most common USB connector. USB 3.0 brought significant advancements to USB connectors.

Purpose

Upon completing this project, you will better understand the purpose of USB-A, Mini-A, Micro-A connectors, and USB 3.0.

Steps for Completion

1. List at least two devices that typically use USB-A connectors.
    a. _____
    _____
2. List at least two devices that typically use Micro-A or Mini-A connectors.
    a. _____
    _____
3. What is one improvement made to USB connections with USB 3.0?
    a. _____
    _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: VGA, USB, and Thunderbolt
Subtopic: USB-A

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.5 USB-A, mini-A, micro-A, 3.0

USB-B, Mini-B, Micro-B, and USB 3.0

USB-B and its variants—Mini-B and Micro-B—are designed to connect peripherals such as printers, external drives, and mobile devices to computers. These connectors evolved to provide better performance, faster speeds, and improved power delivery.

Purpose

Upon completing this project, you will better understand the purpose of USB-B, Mini-B, Micro-B, and USB 3.0.

Steps for Completion

1. List at least two devices that typically use USB-B connectors.
    a. _____
    _____
2. List two devices that typically use Mini-B connectors.
    a. _____
    _____
3. List at least two devices that typically use Micro-B connectors.
    a. _____
    _____
4. What speeds does USB 2.0 support?
    a. _____
5. What speeds does USB 3.0 support?
    a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: VGA, USB, and Thunderbolt
Subtopic: USB-B

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.6 USB-B, mini-B, micro-B, 3.0

USB-C

USB-C is the most advanced and versatile USB standard, designed to replace USB-A, USB-B, and older Micro and Mini connectors with a single universal solution. USB-C offers faster data transfer, higher power delivery, and a reversible connector design that eliminates orientation issues.

Purpose

Upon completing this project, you will better understand the purpose of USB-C.

Steps for Completion

1. What three functions can USB-C handle simultaneously?
    a. _____
    _____
2. What speeds does USB 3.1 support?
    a. _____
3. What speeds does USB 3.2 support?
    a. _____
4. What speeds does USB4 support?
    a. _____
5. How much power can USB-C provide to devices?
    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: VGA, USB, and Thunderbolt
Subtopic: USB-C

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.7 USB-C

Thunderbolt

Thunderbolt is a high-speed connectivity standard developed by Intel, combining data transfer, video output, and power delivery into a single USB-C connector. Thunderbolt guarantees universal compatibility across certified devices.

Purpose

Upon completing this project, you will better understand the purpose of Thunderbolt.

Steps for Completion

1. What speeds does Thunderbolt 4 support?
    a. _____
2. What minimum PCIe data transfer speeds does Thunderbolt 4 support?
    a. _____
3. How many downstream ports with hubs does Thunderbolt 4 support?
    a. _____
4. What is the purpose of Intel VT-d-based DMA protection?
    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: VGA, USB, and Thunderbolt
Subtopic: Thunderbolt

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.8 Thunderbolt

Domain 2

Lesson 6

IT Specialist

Device Configuration and Management

LK LearnKey

S/PDIF

Sony/Philips Digital Interface Format (S/PDIF) is a standardized digital audio transmission technology that allows high-quality sound transfer between devices. It supports optical (TOSLINK) and coaxial connections, transmitting uncompressed stereo or compressed multi-channel audio, such as Dolby Digital and DTS, without interference or signal degradation.

Purpose

Upon completing this project, you will better understand the purpose of Sony/Phillips Digital Interface Format (S/PDIF).

Steps for Completion

1. How does S/PDIF preserve audio fidelity?

    a. _____

2. What type of cable prevents electromagnetic interference with optical S/PDIF?

    a. _____

3. What type of cable prevents electromagnetic interference with coaxial S/PDIF?

    a. _____

4. What is one way S/PDIF is commonly used?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Audio and Conversion Connectors
Subtopic: S/PDIF Optical

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.9 S/PDIF Optical

AUX Cable

The 3.5mm auxiliary (AUX) audio cable is a widely used analog connection for transmitting audio between devices. This versatile cable allows users to connect smartphones, tablets, laptops, and MP3 players to external audio devices such as speakers, car stereos, and headphones. It is a simple plug-and-play solution that does not require additional software or drivers.

Purpose

Upon completing this project, you will better understand the purpose of AUX cables.

Steps for Completion

1. List at least two devices that typically use AUX cables.
    a. _____
    _____
2. What is one upside to using AUX cables for audio transmissions?
    a. _____
    _____
    _____
    _____
    _____
3. What is one downside to using AUX cables for audio transmissions?
    a. _____
    _____
    _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Audio and Conversion Connectors
Subtopic: AUX Audio Cable

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.10 Aux audio cable

LK LearnKey

HDMI Converters

Converting between different audio and video connection types is essential for ensuring compatibility between modern and legacy devices. HDMI converters are key in bridging the gap between devices with varying input and output formats, enabling seamless integration in home entertainment, gaming, and professional setups. HDMI converters translate signals between different formats, allowing users to connect devices that may not share the same port type.

Purpose

Upon completing this project, you will better understand the purpose of HDMI converters.

Steps for Completion

1. How do HDMI converters function?
    a. _____
2. What item may be necessary when using HDMI-to-DVI adapters? Why?
    a. _____
3. What converter can be used to connect newer laptops, tablets, or smartphones to HDMI displays?
    a. _____
4. What two issues can occur with analog conversions?
    a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: Audio and Conversion Connectors
Subtopic: Connection Type Conversions

Objectives covered
2 Features, Apps, and Peripherals
2.4 Compare and contrast capabilities of peripheral connection types
2.4.11 Converting between the various connection types

Domain 2

Lesson 7

IT Specialist

Device Configuration and Management

Wireless Casting

Wireless casting allows users to project their screen from one device to another without using cables, providing a seamless way to share content across different displays. Wireless casting in Windows 11 provides a flexible and efficient way to share content between devices, eliminating the need for cables.

Purpose

Upon completing this project, you will better understand how to use wireless casting.

Steps for Completion

1. What must both sending and receiving devices support to successfully cast content wirelessly?

   a. _____
   _____
   _____

2. What feature must be installed if the receiving device is a Windows 11 PC?

   a. _____

3. What is screen mirroring?

   a. _____

4. What is projection?

   a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: Projection and Display Properties
Subtopic: Wireless Casting

Objectives covered
2 Features, Apps, and Peripherals
2.5 Configure projection and display properties
2.5.1 Wireless casting

Display Orientation

Display orientation refers to the way a screen's content is rotated and displayed based on the physical positioning of the device. It is particularly useful when adjusting external monitors, setting up vertical workspaces for tasks like coding or reading, or adapting tablets and touchscreens for different use cases. Windows 11 provides built-in settings that allow users to change display orientation easily. Windows 11 has four display orientations: Landscape, Portrait, Landscape (Flipped), and Portrait (Flipped).

Purpose

Upon completing this project, you will better understand how to adjust a display's orientation.

Steps for Completion

1. Match the display orientation to its description.

A. Landscape	B. Portrait

   a. _____ The default setting for most desktop monitors, laptops, and televisions, providing a wide viewing area suitable for everyday tasks such as web browsing, office work, and watching videos

   b. _____ Rotates the screen vertically, making it more practical for reading, coding, or working with long documents

2. When might users choose to use the flipped display orientations?

   a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Projection and Display Properties
Subtopic: Orientation

Objectives covered
2 Features, Apps, and Peripherals
2.5 Configure projection and display properties
2.5.2 Orientation

Duplicating and Extending

When using multiple monitors in Windows 11, users can choose between two primary display modes: duplicating and extending. Each mode serves a different purpose depending on how a user wants to interact with multiple screens.

Purpose

Upon completing this project, you will better understand the purpose of duplicating and extending screens.

Steps for Completion

1. What happens when a display is duplicated?
    a. _____
2. What happens when a display is extended?
    a. _____
3. When is it ideal to duplicate a display?
    a. _____
4. When is it ideal to extend a display?
    a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: Projection and Display Properties
Subtopic: Duplicating vs. Extending

Objectives covered
2 Features, Apps, and Peripherals
2.5 Configure projection and display properties
2.5.3 Duplicating vs. extending

Resolution and Aspect Ratio

Resolution and aspect ratio are critical factors in optimizing the display quality of monitors, televisions, and projectors, and understanding how these settings work helps users ensure the best possible visual experience, whether for general use, gaming, video editing, or professional applications. Users can adjust the resolution and aspect ratio in Windows 11.

Purpose

Upon completing this project, you will better understand resolution and aspect ratio.

Steps for Completion

1. What is resolution?

    a. _____
    _____
    _____

2. What is aspect ratio?

    a. _____
    _____
    _____

3. What is the most common aspect ratio?

    a. _____

4. Match the resolution to its description.

A. HD	B. Full HD	C. 4K	D. 8K

    a. _____ 3840 x 2160

    b. _____ 7680 x 4320

    c. _____ 1280 x 720

    d. _____ 1920 x 1080

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 2
Topic: Projection and Display Properties
Subtopic: Resolution and Aspect Ratio

Objectives covered
2 Features, Apps, and Peripherals
2.5 Configure projection and display properties
2.5.4 Resolution and aspect ratio

ClearType

ClearType is a text-rendering technology designed to improve the readability of text on LCD and digital screens. By adjusting how individual pixels display text characters, ClearType enhances sharpness and contrast, reducing eye strain and making text easier to read, especially on lower-resolution displays.

Purpose

Upon completing this project, you will better understand the purpose of ClearType.

Steps for Completion

1. How does ClearType improve the readability of text?

   a. _____

2. On which two types of displays is ClearType most effective?

   a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 2
Topic: Projection and Display Properties
Subtopic: ClearType

Objectives covered
2 Features, Apps, and Peripherals
2.5 Configure projection and display properties
2.5.5 ClearType

Domain 3

Lesson 1

IT Specialist

Device Configuration and Management

Cloud Services

Cloud services, such as OneDrive, Google Drive, and Dropbox, reduce reliance on physical storage devices, ensuring user files remain secure and accessible from anywhere. These services allow users to synchronize files across multiple devices, providing remote access to documents, photos, and videos from any internet-connected device.

Purpose

Upon completing this project, you will become more familiar with Cloud Services.

Steps for Completion

1. What are two key benefits of cloud storage?
    a. _____
2. Label the following statements as true or false.
    a. _____ Files saved in OneDrive can be accessed from any device with the same Microsoft account.
    b. _____ When a file is shared on OneDrive, recipients can always edit it.
    c. _____ OneDrive allows users to restore previous versions of files if necessary.
    d. _____ Most cloud storage platforms provide encryption for files, but do not provide backup solutions.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 3
Topic: Cloud Services
Subtopic: Cloud Storage and Collaboration

Objectives covered
3 Data Access and Management
3.1 Describe cloud services
3.1.1 Cloud storage and collaboration concepts

LK LearnKey

Storage and Service Providers

Various cloud service providers offer storage, productivity, and computing power solutions. Cloud computing has allowed businesses and individuals to manage their data, applications, and collaboration tools remotely. Four of the most widely used cloud service providers are Microsoft Azure, Microsoft 365, Google Workspace, and Amazon Web Services (AWS).

Purpose

Upon completing this project, you will become more familiar with storage and service providers.

Steps for Completion

1. What are two cloud services offered by Microsoft Azure?

   a. _____
   _____
   _____

2. What are two cloud-based productivity tools included in the Microsoft 365 suite?

   a. _____
   _____

3. What are two cloud-based tools offered by Google Workspace?

   a. _____

4. Amazon _____ is a cloud storage solution offered by Amazon Web Services (AWS).

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 3
Topic: Cloud Services
Subtopic: Storage and Service Providers

Objectives covered
3 Data Access and Management
3.1 Describe cloud services
3.1.2 Identify common cloud storage and service providers, such as Azure, Microsoft 365 (to include but not limited to: SharePoint, OneDrive, Outlook, Teams, Windows 365)

File Sharing and Synchronization

Cloud storage services allow users to collaborate in real-time through powerful file-sharing capabilities. Using offline file synchronization, users can access and edit cloud-stored files without an active internet connection. Microsoft OneDrive, SharePoint, Google Drive, and Dropbox services allow seamless sharing and offline file access.

Purpose

Upon completing this project, you will better understand file sharing and synchronization.

Steps for Completion

1. Match each shared content permission to its description.

A. View-only access	C. Owner access
B. Edit access	D. Download and print restrictions

   a. _____ Allows the original user to manage permissions, revoke access, or transfer ownership of files

   b. _____ Allows users to read but not modify files

   c. _____ Allows users to make real-time changes to files

   d. _____ Offered by some platforms to prevent unauthorized file distribution

2. Label the following statements as true or false.

   a. _____ Offline file synchronization means changes to files will sync automatically once a device reconnects to the internet.

   b. _____ In OneDrive, any changes made to files while offline will need to be synced manually when an internet connection is restored.

3. If OneDrive is syncing properly, a _____ checkmark icon appears next to a locally stored file.

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 3
Topic: Cloud Services
Subtopic: File Sharing; Offline File Synchronization

Objectives covered
3 Data Access and Management
3.1 Describe cloud services
3.1.3 File sharing capabilities and permissions
3.1.4 Offline file synchronization

Domain 3

Lesson 2

IT Specialist

Device Configuration and Management

LK LearnKey

Partition and Format a Drive

Preparing a storage device for use involves partitioning and formatting it. Partitioning means dividing a physical storage device into multiple sections, and formatting means preparing a partition for storing data by setting up a file system.

Purpose

Upon completing this project, you will better understand partitioning and formatting a drive.

Steps for Completion

1. What is one benefit of partitioning a physical disk?

    a. _____
    _____
    _____

2. Label the following statements as true or false.

    a. _____ Partitioning a storage device benefits users who are managing large amounts of data, running multiple operating systems, or requiring dedicated partitions for backup purposes.

    b. _____ Partitioning improves security by keeping personal files and system files together.

    c. _____ Partitioning can facilitate multi-OS installations, allowing Windows and Linux to run independently.

3. Which file system is used for compatibility across different operating systems?

    a. _____

4. Which file system is for Windows-only use?

    a. _____

5. Which Windows tool is used to partition and format a hard drive?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 3
Topic: Storage
Subtopic: Partition and Format a Drive

Objectives covered
3 Data Access and Management
3.2 Describe and configure storage
3.2.1 Understand when and why to use partitioning
3.2.2 Partition and format a drive
3.2.3 Choose a file system to use when formatting the drive (NTFS, FAT32, and exFAT)

GPT and MBR Partition Styles

Users must choose between the GUID Partition Table (GPT) and the Master Boot Record (MBR) styles when partitioning a new drive in Windows 11. Users who understand when to choose which style ensure efficient storage management and compatibility with an operating system.

Purpose

Upon completing this project, you will better understand GPT and MBR partition styles.

Steps for Completion

1. Determine whether each statement refers to the GPT or MBR partition style.
    a. _____ Limited to four partitions and a 2TB disk
    b. _____ Recommended for modern systems using UEFI
    c. _____ Supports disks over 2TB and more than four partitions
    d. _____ Supports older systems that require BIOS mode
2. Label the steps for formatting a drive in the correct order from 1 to 5.
    a. _____ Format the partition
    b. _____ Initialize the disk
    c. _____ Set a file system
    d. _____ Assign a drive letter
    e. _____ Choose the partition style

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 3
Topic: Storage
Subtopic: GPT and MBR Partition Style

Objectives covered
3 Data Access and Management
3.2 Describe and configure storage
3.2.6 GPT and MBR partition style

Domain 3

Lesson 3

IT Specialist

Device Configuration and Management

LK LearnKey

File, Share, and Effective Permissions

In Windows 11, file and share permissions ensure only authorized users can access and interact with system resources. These permissions are essential for controlling access, modification, or management of files and folders. Effective permissions determine a user or group's final access rights to a file or folder.

Purpose

Upon completing this project, you will become more familiar with file and share permissions.

Steps for Completion

1. Match each file permission to its description.

A. Read	C. Modify
B. Write	D. Full Control

   a. _____ Allows users to read, write, and delete files but prevents them from changing security settings

   b. _____ Allows users to view file contents without making changes

   c. _____ Provides unrestricted access to users, enabling them to modify, delete, and adjust permissions for other users

   d. _____ Allows users to modify content but not delete or change permissions

2. _____ permissions are applied at the local level, while _____ permissions come into effect when a file or folder is accessed over a network.

3. _____ permissions result from a combination of explicit and inherited permissions.

4. _____ permissions are assigned directly to a user, while _____ permissions come from the parent folder.

5. Which share permission allows users to edit and delete content?

   a. _____

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 3
Topic: Local and Network File Permissions
Subtopic: File and Share Permissions; Effective Permissions

Objectives covered
3 Data Access and Management
3.3 Describe and configure local and network file sharing and permissions
3.3.1 File and share permissions
3.3.2 Effective permissions

Basic and Advanced Permissions

Two levels of permission control are available in Windows 11: basic and advanced. Windows 11 also provides multiple ways to share files across a network, with Public, Basic, and Advanced Shares providing different levels of access control.

Purpose

Upon completing this project, you will better understand basic and advanced permissions and shares.

Steps for Completion

1. What is the difference between basic and advanced permissions?

   a. _____
   _____
   _____

2. Determine whether each permission is basic (B) or advanced (A).

   a. _____ Full Control

   b. _____ Traverse Folder/Execute File

   c. _____ Modify

   d. _____ Read & Execute

   e. _____ Delete Subfolders

   f. _____ List Folder Contents

   g. _____ Read

   h. _____ Change Permissions

   i. _____ Create Files/Write Data

   j. _____ Write

3. _____ shares make files accessible to all users on a network, making it easier to collaborate at the cost of security controls. These shares are ideal for environments where accessibility is more important than restricting access.

4. _____ shares provide detailed control over permissions, such as setting user-specific access, creating hidden shares, and applying security auditing.

5. _____ shares allow users to specify who can access shared folders and define their level of interaction. These shares are most suitable for environments where only specific users need access.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 3
Topic: Local and Network File Permissions
Subtopic: Basic and Advanced Permissions; Public, Basic, and Advanced Shares

Objectives covered
3 Data Access and Management
3.3 Describe and configure local and network file sharing and permissions
3.3.3 Basic and advanced permissions
3.3.4 Public, basic, and advanced shares

LearnKey

Mapping a Drive and Taking Ownership

Mapping a network drive makes accessing network resources easier and improves workflow efficiency. Taking ownership of a file or folder is necessary when permission settings restrict access.

Purpose

Upon completing this project, you will become more familiar with mapping a drive and taking ownership of a file or folder.

Steps for Completion

1. How does one map a network drive?
    a. _____
2. Once a drive is mapped in Windows, where can one find it?
    a. _____
3. When might taking ownership of a file be useful?
    a. _____
4. What is the purpose of running the takeown /f "C:\Path\To\File" /r/d y command?
    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 3
Topic: Local and Network File Permissions
Subtopic: Map Drives; Taking Ownership

Objectives covered
3 Data Access and Management
3.3 Describe and configure local and network file sharing and permissions
3.3.5 Map drives
3.3.6 Describe taking ownership of files or folders

Domain 3

Lesson 4

IT Specialist

Device Configuration and Management

LearnKey

Backup and Restore

Windows 11 offers various tools for managing the backup and restoration of user files and states. File Version and History settings are essential for preserving multiple file versions. A full disk backup ensures that an entire system is securely stored and recoverable, while a full disk restore is used to recover a previously created backup. Restoring previous versions of files allows users to recover documents, images, or other files that have been unintentionally modified, deleted, or corrupted.

Purpose

Upon completing this project, you will better understand backup and restore tools in Windows 11.

Steps for Completion

1. What does File History do?

   a. _____
   _____
   _____

2. Label the following statements as true or false.

   a. _____ Full disk backup includes the operating system, applications, settings, and personal files.

   b. _____ Windows 11 includes a built-in feature for full disk backups directly to the cloud.

   c. _____ Full disk restore is used to recover an entire system from a previously created backup, returning it to a functional state in case of corruption, failure, or hardware replacement.

   d. _____ System Restore can revert system settings and files to a previous state for broader system rollbacks than restoring previous versions of files.

   e. _____ Restoring a file will always overwrite the current file.

   f. _____ System Restore will affect files and folders when restoring system settings.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 3
Topic: Backup and Restore
Subtopic: File Versioning and History; Full Disk Backup to the Cloud; Full Disk Restore; Restore Previous Versions of Files

Objectives covered
3 Data Access and Management
3.4 Manage backup and restore of user files and states
3.4.1 Set file versioning/history Settings
3.4.2 Perform a full disk backup to the cloud
3.4.3 Perform a full disk restore
3.4.4 Restore previous versions of files

Domain 4

Lesson 1

IT Specialist

Device Configuration and Management

Allow an App or Feature

Windows Defender Firewall protects a system from viruses and other malware. Sometimes, it will block a legitimate app or feature someone needs for work. When this happens, that specific feature can be allowed through Windows Defender Firewall.

Purpose

Upon completing this project, you will better understand how to allow an app or feature through Windows Defender Firewall.

Steps for Completion

1. Open Windows Defender Firewall.
2. Select Allow an app or feature through Windows Defender Firewall.
3. Take a screenshot of the window that appears.
4. Save the screenshot as **411-Completed.png** to your Domain 4 Student folder.
5. How can one ensure access to needed apps without compromising security on a public network?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Windows Defender Firewall
Subtopic: Allow an Application or Feature

Objectives covered
4 Device Security
4.1 Configure Windows Defender Firewall settings
4.1.1 Allow an application or feature through the Windows Defender Firewall

Network Profiles

Windows 11 offers different network profiles to manage how a device interacts with other devices on the same network. Selecting the appropriate profile ensures a balance between security and connectivity. The available profile types are private, public, and guest.

Purpose

Upon completing this project, you will better understand how to use network profiles.

Steps for Completion

1. Describe each of the following network profile types.
    a. Private: _____
    _____
    _____
    b. Public: _____
    _____
    _____
    _____
    c. Guest: _____
    _____
    _____
    _____
    _____
2. Label each location with either Private or Public, for the network profile type that should be used in each.
    a. Coffee shop: _____
    b. Office: _____
    c. Hotel: _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Windows Defender Firewall
Subtopic: Network Profiles

Objectives covered
4 Device Security
4.1 Configure Windows Defender Firewall settings
4.1.2 Compare and contrast private, public, and guest network profiles

Firewall Troubleshooting

Windows Defender Firewall protects a system from unauthorized access, but in some cases, temporarily disabling it can help troubleshoot connectivity issues. However, this should only be done briefly to prevent security vulnerabilities.

Purpose

Upon completing this project, you will better understand how to turn a firewall on and off for troubleshooting purposes.

Steps for Completion

1. Label the following statements as true or false.
    a. _____ Windows Defender Firewall can be turned off if a third-party firewall is running instead.
    b. _____ It is acceptable to leave the firewall turned off if it is consistently causing issues accessing needed features.
2. Go into Windows Defender Firewall through settings and practice turning it off and on. Make sure it is turned on when you have finished.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic:Windows Defender Firewall
Subtopic: Firewall Troubleshooting

Objectives covered
4 Device Security
4.1 Configure Windows Defender Firewall settings
4.1.3 Turn firewall off and on (for troubleshooting)

Domain 4

Lesson 2

IT Specialist

Device Configuration and Management

LearnKey

Multifactor Authentication

Multifactor authentication (MFA) enhances security by requiring users to provide multiple forms of verification to access an account or system. This added layer of protection helps prevent unauthorized access, even if a password is compromised. By implementing MFA, users and organizations can significantly reduce the risk of unauthorized access and strengthen security measures.

Purpose

Upon completing this project, you will better understand multifactor authentication.

Steps for Completion

1. List an example for each authentication factor type.
    a. Something You Know: _____
    _____
    b. Something You Have: _____
    _____
    c. Something You Are: _____
    _____
2. What are two reasons many organizations enforce MFA?
    a. _____
    _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: User Authentication Options
Subtopic: Multifactor Authentication

Objectives covered
4 Device Security
4.2 Describe user authentication and configure Windows sign-in options
4.2.1 Multifactor authentication (theory, how it works)

Biometric Authentication

Biometric authentication enhances security by verifying a user's identity using unique physical or behavioral traits. Unlike passwords, which can be stolen or forgotten, biometrics provide a convenient and secure authentication method.

Purpose

Upon completing this project, you will better understand biometric authentication methods.

Steps for Completion

1. List three biometric authentication methods.
    a. _____
    _____
    _____
2. Label the following statements as true or false.
    a. _____ Fingerprint authentication only works for devices with the corresponding hardware.
    b. _____ Biometric authentication does not need to be paired with additional security measures such as MFA.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: User Authentication Options
Subtopic: Biometric Authentication Methods

Objectives covered
4 Device Security
4.2 Describe user authentication and configure Windows sign-in options
4.2.2 Biometric authentication methods

Windows Sign-In Options

Windows 11 provides multiple sign-in options to enhance security and convenience, allowing users to select the authentication method that best fits their needs. Properly configuring these options ensures secure access to accounts while reducing reliance on traditional passwords, which are vulnerable to hacking attempts.

Purpose

Upon completing this project, you will better understand how to configure Windows Hello and other sign-in options.

Steps for Completion

1. What is Windows Hello?
    a. _____
2. What are three sign-in options available for Windows besides Windows Hello?
    a. _____
3. Under which section of settings can sign-in options be found?
    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: User Authentication Options
Subtopic: Windows Hello; Configure Windows Sign-In Options

Objectives covered
4 Device Security
4.2 Describe user authentication and configure Windows sign-in options
4.2.3 Windows Hello (Windows 11 only)
4.2.4 Configure Windows sign-in options

LK LearnKey

Strong Passwords

Passwords are the first defense against unauthorized access, but weak passwords remain one of the most common causes of security breaches. A strong password significantly reduces the risk of cyberattacks by making it more difficult for hackers to guess or crack. To further protect accounts, multifactor authentication (MFA) should be enabled whenever possible, ensuring that even if a password is compromised, unauthorized access is still prevented.

Purpose

Upon completing this project, you will better understand what makes a password strong.

Steps for Completion

1. Label the following statements as true or false.

    a. _____ It is essential to avoid common words, predictable sequences, and personal information when creating a password.

    b. _____ A strong password must be at least 6–10 characters long and include a mix of uppercase and lowercase letters, numbers, and special characters.

    c. _____ A password manager is a good solution for generating and storing strong passwords for multiple sites.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: User Authentication Options
Subtopic: Strong Passwords

Objectives covered
4 Device Security
4.2 Describe user authentication and configure Windows sign-in options
4.2.5 What makes a password "strong"

LK LearnKey

Authenticator Apps

Authenticator apps are one method of generating codes for multifactor authentication. They are more secure than SMS-based authentication and provide strong protection when paired with a password or biometric authentication method.

Purpose

Upon completing this project, you will better understand authenticator apps.

Steps for Completion

1. Why are authenticator apps more secure than SMS authentication?

    a. _____
    _____
    _____

2. Label the following statements true or false.

    a. _____ Apps can be downloaded through the App Store.

    b. _____ An authenticator app can only be linked to one account.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: User Authentication Options
Subtopic: Authenticator Apps

Objectives covered
4 Device Security
4.2 Describe user authentication and configure Windows sign-in options
4.2.6 Authenticator apps

Domain 4

Lesson 3

IT Specialist

Device Configuration and Management

LK LearnKey

Computer Viruses

A computer virus is malicious software (malware) that infects a device by embedding itself into programs, files, or system processes. Once executed, a virus can replicate, spread to other files or devices, and perform harmful actions such as deleting data, corrupting files, stealing information, or even making the system inoperable. Viruses typically require user action to spread, such as opening an infected email attachment, downloading unverified software, or inserting an infected USB drive.

Purpose

Upon completing this project, you will better understand different computer viruses and how to respond to them.

Steps for Completion

1. Match each virus to its description.

A.	File Infector	C.	Macro	E.	Multipartite
B.	Boot Sector	D.	Polymorphic		

   a. _____ Viruses infect the boot sector of a hard drive or removable media, such as USB drives

   b. _____ Viruses that infect multiple parts of a system simultaneously, such as the boot sector, files, and memory

   c. _____ Viruses that are embedded in documents, such as Microsoft Word or Excel files, and can activate when a document is opened

   d. _____ Viruses that constantly change their code to evade detection by antivirus software, making them more challenging to remove

   e. _____ Viruses that attach themselves to executable files (.exe, .dll) and activate when the infected program is opened

2. What are three ways to minimize the risk of infection by computer viruses?

   a. _____
   _____
   _____

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 4
Topic: Various Security Threats
Subtopic: Computer Viruses

Objectives covered
4 Device Security
4.3 Describe various security threats
4.3.1 Computer viruses (worms, trojan horse, logic bombs, ransomware)
4.4 Describe how to respond to various malware and social engineering attacks
4.4.1 Computer viruses

Adware

Adware (advertising-supported software) is malware designed to generate revenue through intrusive advertisements. While not always directly harmful, adware slows down system performance, tracks user activity, and floods the screen with pop-ups, banners, or redirect links. Some adware can expose users to more serious threats, such as phishing sites, spyware, or ransomware.

Purpose

Upon completing this project, you will better understand adware and how to prevent it.

Steps for Completion

1. What are two ways adware can enter a system?

   a. _____

2. What are three possible negative effects of adware?

   a. _____

3. What are three ways adware infections can be prevented?

   a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Various Security Threats
Subtopic: Adware

Objectives covered
4 Device Security
4.3 Describe various security threats
4.3.2 Adware
4.4 Describe how to respond to various malware and social engineering attacks
4.4.2 Adware

LK LearnKey

Spyware

Spyware is malware that secretly collects information from a device without a user's knowledge. It operates in the background, tracking keystrokes, sign-in credentials, browsing history, and personal information. Spyware is often used for identity theft, financial fraud, or corporate espionage.

Purpose

Upon completing this project, you will better understand spyware and how to prevent it.

Steps for Completion

1. Match each spyware type to its description.

A.	Keylogger	C.	Screen scrapers
B.	Browser hijackers	D.	Trojan

   a. _____ Modify browser settings, change homepages, inject advertisements, and track user activity

   b. _____ Disguised as legitimate software, it installs backdoors that allow attackers to access sensitive files and data

   c. _____ Records every keystroke a user types, capturing passwords, credit card details, and private messages

   d. _____ Take screenshots of a user's activity, including online banking transactions and secure sign-in pages

2. What are three ways to protect against spyware?

   a. _____
   _____
   _____
   _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Various Security Threats
Subtopic: Spyware

Objectives covered
4 Device Security
4.3 Describe various security threats
4.3.3 Spyware
4.4 Describe how to respond to various malware and social engineering attacks
4.4.3 Spyware

LK LearnKey

Denial of Service (DoS) Attacks

A Denial of Service (DoS) attack is a malicious attempt to disrupt the normal operation of a server, network, or online service by overwhelming it with excessive traffic. A Distributed Denial of Service (DDoS) attack is a more advanced form of DoS, where multiple compromised devices, often forming a botnet, are used to flood a target with traffic, making it significantly harder to mitigate. These attacks prevent legitimate users from accessing services, causing downtime, financial loss, and reputational damage.

Purpose

Upon completing this project, you will better understand DoS attacks and how to prevent them.

Steps for Completion

1. Describe each type of DoS attack.
    a. Volume-Based: _____
    _____
    _____
    b. Protocol: _____
    _____
    c. Application Layer: _____
    _____
2. What are two potential consequences of a successful DoS or DDoS attack?
    a. _____
    _____
3. What are three ways to mitigate the risk of DoS and DDoS attacks?
    a. _____
    _____
    _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Various Security Threats
Subtopic: Denial of Service (DoS) Attacks

Objectives covered
4 Device Security
4.3 Describe various security threats
4.3.4 Denial of Service (DoS) attacks

Social Engineering Attacks

Social engineering attacks manipulate human psychology to trick individuals into revealing sensitive information, granting unauthorized access, or performing actions compromising security. Unlike cyberattacks that exploit software vulnerabilities, these attacks rely on deception, trust, and urgency to bypass security defenses. They often occur via emails, phone calls, text messages, or in-person interactions.

Purpose

Upon completing this project, you will better understand social engineering attacks and how to avoid them.

Steps for Completion

1. Describe each type of social engineering attack.
    a. Pretexting: _____
    b. Baiting: _____
    c. Quid pro quo scheme: _____
    d. Tailgating: _____
    e. Dumpster diving: _____
    f. Spoofing: _____
    g. Phishing: _____
    h. Smishing: _____
    i. Vishing: _____
2. What are three ways to prevent social engineering attacks?
    a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 4
Topic: Various Security Threats
Subtopic: Social Engineering Attacks

Objectives covered
4 Device Security
4.3 Describe various security threats
4.3.5 Social engineering attacks (to include, but not limited to Phishing, smishing, vishing, dumpster diving, spoofing, and clone phishing)
4.4 Describe how to respond to various malware and social engineering attacks
4.4.4 Phishing

Physical Attacks

Cyber-physical threats are growing as attackers exploit the intersection of digital and physical security vulnerabilities. Unlike traditional cyberattacks, these threats target digital systems and physical infrastructure, posing risks to individuals, businesses, and critical industries. As more devices connect through the Internet of Things (IoT), attackers have more opportunities to exploit physical weaknesses to gain digital access.

Purpose

Upon completing this project, you will better understand physical attacks and how to prevent them.

Steps for Completion

1. List four examples of cyber-physical threats.

    a. _____
    _____

2. What are three ways to mitigate physical threats?

    a. _____
    _____
    _____
    _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Various Security Threats
Subtopic: Physical Attacks

Objectives covered
4 Device Security
4.3 Describe various security threats
4.3.6 Physical attacks (errant thumb drives, theft, shoulder surfing, screen scrapers)
4.4 Describe how to respond to various malware and social engineering attacks
4.4.5 Physical attacks **(**errant thumb drives**)**

Domain 4

Lesson 4

IT Specialist

Device Configuration and Management

LearnKey

Antimalware App Options

Windows Security, formerly known as Windows Defender, provides built-in protection that, when properly configured, ensures a system remains secure while minimizing performance disruptions. Windows Security offers real-time protection, automatic updates, and multiple scanning options to defend against threats like trojans, worms, adware, and ransomware.

Purpose

Upon completing this project, you will better understand antimalware app options.

Steps for Completion

1. What is the benefit of scheduling automatic scans?

    a. _____

2. What are two ways of ensuring further security using an antimalware app?

    a. _____

3. When should exclusions be configured in Windows security?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 4
Topic: Antimalware Options and Results
Subtopic: Antimalware App Options

Objectives covered
4 Device Security
4.4 Describe how to respond to various malware and social engineering attacks
4.4.6 Antimalware program configuration options

LearnKey

Antimalware Scan Results

Understanding how to analyze scan results is essential for responding appropriately to detected threats. After running a scan in Windows Security, the summary will display details on detected threats, categorized by severity and location. Threats are classified as either low, medium, or high severity.

Purpose

Upon completing this project, you will better understand how to analyze antimalware scan results.

Steps for Completion

1. Label each threat as either high, medium, or low severity.
    a. Ransomware: _____
    b. PUPs: _____
    c. Spyware: _____
    d. Trojans: _____
    e. Adware: _____
2. Explain each of the response options available in Windows Security.
    a. Quarantine: _____
    _____
    b. Remove: _____
    c. Restore: _____
    _____
    d. Submit for further analysis: _____
    _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 4
Topic: Antimalware Options and Results
Subtopic: Analyze Antimalware Scan Results

Objectives covered
4 Device Security
4.4 Describe how to respond to various malware and social engineering attacks
4.4.7 Analyze Antimalware scan results

Domain 4

Lesson 5

IT Specialist

Device Configuration and Management

LK LearnKey

User Account Control (UAC)

User Account Control (UAC) is a critical security feature in Windows that helps prevent unauthorized changes to a system by requiring user approval before allowing applications or system settings to make significant modifications. This security measure ensures that malware or unauthorized users cannot silently alter system configurations without explicit permission.

Purpose

Upon completing this project, you will better understand the purpose of UAC.

Steps for Completion

1. What are three events that will trigger a UAC dialog box asking for confirmation before proceeding?

   a. _____
   _____
   _____

2. What is the most secure UAC setting?

   a. _____

3. Which UAC setting is ideal for home users who install software, browse the internet, and manage personal files?

   a. _____

4. Which UAC setting is ideal for shared computers used by multiple people?

   a. _____

5. Configuring _____ always to run as an administrator ensures IT professionals can execute scripts without manually elevating permissions each time.

6. For users managing multiple UAC permissions, the _____ Group Policy Editor allows advanced configuration of elevation behavior.

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 4
Topic: UAC Settings
Subtopic: Describe the Function of UAC; UAC Setting Levels; Elevate Permissions in UAC

Objectives covered
4 Device Security
4.5 Manage User Account Control (UAC) settings
4.5.1 Describe the function of UAC
4.5.2 Identify appropriate UAC settings for specific purposes
4.5.3 Elevate permissions in UAC

Domain 5

Lesson 1

IT Specialist

Device Configuration and Management

LK LearnKey

Windows Troubleshooting Tools

Windows 11 provides various troubleshooting tools to help users diagnose and resolve system issues. Knowing how to locate and use these tools ensures that users can monitor system health and performance effectively.

Purpose

Upon completing this project, you will become more familiar with Windows troubleshooting tools.

Steps for Completion

1. Open Event Viewer in Windows 11.
2. Open the Application log and double-click an event listed there.
3. Take a screenshot and save the file as **511a-Completed.png** to your Domain 5 Student folder.
4. Open Task Manager and view the Processes tab.
5. Take a screenshot of the dialog box.
6. Save the file as **511b-Completed**
7. Open the Defragment and Optimize Drives tool.
8. Take a screenshot of the dialog box.
9. Save the file as **511c-Completed**
10. What operation does Windows 11 automatically perform on SSDs instead of traditional defragmentation?
    a. _____
11. What is the keyboard shortcut to open Event Viewer?
    a. _____
12. What is the keyboard shortcut to open Task Manager?
    a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 5
Topic: Troubleshooting Tasks
Subtopic: Windows Troubleshooting Tools

Objectives covered
5 Management and Troubleshooting
5.1 Perform troubleshooting tasks
5.1.1 Locate and identify Windows troubleshooting tools (such as Event Viewer, Task Manager, defragment and optimize drive)

LK LearnKey

Gather Data on Issues

The right data is crucial when identifying system issues and determining appropriate fixes. Windows 11 provides users with several tools for gathering this information.

Purpose

Upon completing this project, you will better understand gathering data on system issues.

Steps for Completion

1. Match each tool to its corresponding function.

A. Event Viewer	D. PowerShell
B. Task Manager	E. Reliability Monitor
C. System Information	

   a. _____ Details hardware specifications, BIOS versions, and installed drivers

   b. _____ Provides system logs with error messages and failure details

   c. _____ Tracks system stability and past failures

   d. _____ Helps diagnose performance issues caused by resource-heavy applications

   e. _____ Enables advanced troubleshooting by running commands.

2. What does running PowerShell as an administrator allow users to do?

   a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Troubleshooting Tasks
Subtopic: Gather Data on Issues

Objectives covered
5 Management and Troubleshooting
5.1 Perform troubleshooting tasks
5.1.2 Gather data to describe issues and support troubleshooting

LK LearnKey

Research Issue Remedies

Once users have gathered troubleshooting data, the next step is to research potential solutions. This step means exploring reliable sources and is instrumental when basic troubleshooting does not immediately resolve a complex issue.

Purpose

Upon completing this project, you will become more familiar with researching issue remedies.

Steps for Completion

1. What are two reliable sources for researching issue remedies?
    a. _____
    _____
    _____
2. What is one of the first tools to leverage before using online research?
    a. _____
3. Label the following statements as true or false.
    a. _____ Community-driven platforms such as Stack Overflow and Microsoft Community Forums often contain discussions and solutions from users facing similar issues.
    b. _____ Platforms like Reddit and tech blogs should not be trusted to fix Windows problems.
    c. _____ PowerShell commands can be effective for more advanced troubleshooting.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Troubleshooting Tasks
Subtopic: Research Issue Remedies

Objectives covered
5 Management and Troubleshooting
5.1 Perform troubleshooting tasks
5.1.3 Research how to remedy issues

LearnKey

Identify When to Escalate Issues

Sometimes, escalation is necessary when problems persist despite troubleshooting, require administrative access, or impact multiple users or critical systems. Knowing when to escalate is crucial for ensuring a faster and more effective resolution to problems involving physical repairs, affecting business operations, involving IT support, vendors, or security specialists.

Purpose

Upon completing this project, you will better understand how to identify when to escalate issues.

Steps for Completion

1. Before escalating an issue, what should be done to ensure a smooth handoff to the next level of support?

   a. _____

2. To whom should an issue be escalated if a Group Policy restriction prevents access to system settings?

   a. _____

3. To whom should an issue be escalated if third-party software requires access to specialized tools and patches?

   a. _____

4. To whom should the issue be escalated if security-related incidents such as unauthorized sign-in attempts or malware infections occur?

   a. _____

5. What should a detailed support ticket submitted to an IT help desk include?

   a. _____
   _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Troubleshooting Tasks
Subtopic: Identify When to Escalate Issues

Objectives covered
5 Management and Troubleshooting
5.1 Perform troubleshooting tasks
5.1.4 Identify when to escalate issues

Force Group Policy Application

An essential feature in Windows 11 that allows administrators to enforce security settings, software configurations, and user restrictions across multiple devices is Group Policy. Group Policy updates automatically at regular intervals, but sometimes changes need to be applied immediately to take effect by forcing a Group Policy update.

Purpose

Upon completing this project, you will better understand when to force Group Policy application.

Steps for Completion

1. Open the Command Prompt as an Administrator.
2. Run the command to force a Group Policy update manually.
3. Take a screenshot of the result.
4. Save the file as **515a-Completed.png** to your Domain 5 Student folder.
5. Run the command to list all applied policies.
6. Take a screenshot of the result.
7. Save the file as **515b-Completed**
8. What does the command, gpresult /h C:\GPReport.html, do?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 5
Topic: Troubleshooting Tasks
Subtopic: Force Group Policy Application

Objectives covered
5 Management and Troubleshooting
5.1 Perform troubleshooting tasks
5.1.5 Force group policy application (gpupdate /force, gpresult)

LK LearnKey

Problems with Applied Policies

Group Policy is a powerful tool for managing Windows environments, and sometimes misconfigured policies can lead to unintended issues. These issues include preventing user sign-ins, blocking network access, or causing software malfunctions. Recognizing when a Group Policy setting is the source of a problem is important for troubleshooting and maintaining system functionality.

Purpose

Upon completing this project, you will better understand how to troubleshoot applied policy issues.

Steps for Completion

1. What is the first step of diagnosis when a Group Policy setting is suspected to cause an issue?
    a. _____
2. Which Event IDs indicate a failed Group Policy application?
    a. _____
3. How does one temporarily disable a problematic Group Policy?
    a. _____
4. What does the command, secedit /configure /cfg%windir%\inf\deftbase.inf /db deftbase.sdb /verbose, do?
    a. _____
5. Open Event Viewer and check for Group Policy errors.
6. Take a screenshot of the Event Viewer.
7. Save the file as **516-Completed.png** in your Domain 5 Student folder.

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 5
Topic: Troubleshooting Tasks
Subtopic: Problems with Applied Policies

Objectives covered
5 Management and Troubleshooting
5.1 Perform troubleshooting tasks
5.1.6 Recognize that an applied policy could cause a problem

Domain 5

Lesson 2

IT Specialist

Device Configuration and Management

LK LearnKey

Resets and Rollbacks

Resetting or rolling back an operating system allows users to restore the system to a stable state while preserving or removing personal data. Resetting Windows reinstalls the operating system, while rolling back restores a previous version after an update, if the rollback option is available.

Purpose

Upon completing this project, you will become more familiar with Windows resets and rollbacks.

Steps for Completion

1. When should one consider resetting Windows 11?

   a. _____

2. When should one consider rolling back Windows 11?

   a. _____

3. What does the Keep my files option do when resetting a PC?

   a. _____

4. What does the Remove everything option do when resetting a PC?

   a. _____

5. Label the following statements as true or false.

   a. _____ The rollback option in Windows 11 is only available 10 days after an update.

   b. _____ Windows Recovery Environment (WinRE) is accessible in Windows Settings.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Operating Systems and Applications
Subtopic: Resets and Rollbacks

Objectives covered
5 Management and Troubleshooting
5.2 Troubleshoot operating system and application issues
5.2.1 Reset or roll back the operating system

Advanced Startup

Advanced Startup provides access to recovery tools when Windows 11 encounters critical errors preventing normal booting. These tools allow users to restore previous system states, run command-line diagnostics, repair startup files, and reinstall Windows 11 if necessary.

Purpose

Upon completing this project, you will become more familiar with Advanced Startup.

Steps for Completion

1. List three ways to access Advanced Startup.
    a. _____
2. What is the primary tool in Advanced Startup for fixing boot-related issues?
    a. _____
3. What is System Image Recovery?
    a. _____
4. What is the Command Prompt command that scans for and repairs corrupted system files?
    a. _____
5. If a newly installed graphics driver causes Windows to crash, which Advanced Startup tool would be best?
    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Operating Systems and Applications
Subtopic: Advanced Startup

Objectives covered
5 Management and Troubleshooting
5.2 Troubleshoot operating system and application issues
5.2.2 Advanced startup (System repair

Safe Mode

Safe Mode in Windows 11 is a powerful troubleshooting tool that allows the system to start with only essential drivers and services. This diagnostic mode is particularly useful when dealing with software conflicts, malware infections, or driver-related problems that prevent the system from functioning normally. By running Windows in a minimal state, Safe Mode helps isolate the root cause of a problem and enables users to take corrective action.

Purpose

Upon completing this project, you will better understand the purpose of Windows Safe Mode.

Steps for Completion

1. Label the following statements as true or false.

    a. _____ Safe Mode prevents most malware from running.

    b. _____ Safe Mode turns on third-party software and startup applications.

2. Match the Safe Mode variation to its description.

A. Standard Safe Mode	C. Safe Mode with Command Prompt
B. Safe Mode with Networking	

    a. _____ Includes network drivers to allow internet access for troubleshooting or downloading updates

    b. _____ Provides a command-line interface for advanced troubleshooting

    c. _____ Loads only essential system drivers

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Operating Systems and Applications
Subtopic: Safe Mode Features

Objectives covered
5 Management and Troubleshooting
5.2 Troubleshoot operating system and application issues
5.2.3 Features of safe mode

LearnKey

Application Compatibility Issues

Application compatibility issues arise when software designed for older versions of Windows fails to function correctly in Windows 11. These issues can manifest as crashes, incorrect UI behavior, slow performance, or error messages stating that a program is not supported. Identifying and resolving these issues is crucial for maintaining productivity, especially with legacy applications.

Purpose

Upon completing this project, you will better understand how to troubleshoot application compatibility issues.

Steps for Completion

3. What is the purpose of the Compatibility Troubleshooter?

    a. _____

4. What is the purpose of the Program Compatibility Assistant (PCA)?

    a. _____

5. Which Windows tool can be used for advanced troubleshooting to diagnose application failures?

    a. _____

6. What is the purpose of the Windows Compatibility Center?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5-10 minutes

Video reference
Domain 5
Topic: Operating Systems and Applications
Subtopic: Identify App Compatibility Issues

Objectives covered
5 Management and Troubleshooting
5.2 Troubleshoot operating system and application issues
5.2.4 Use troubleshooting tools to identify application compatibility issues

Domain 5

Lesson 3

IT Specialist

Device Configuration and Management

LK LearnKey

Microsoft Store Application Issues

Microsoft Store applications sometimes fail to download, install, or update due to internet connectivity problems, corrupted cache files, insufficient storage, or system misconfigurations. Troubleshooting these issues ensures users can access and use Microsoft Store applications without interruptions.

Purpose

Upon completing this project, you will better understand how to troubleshoot Microsoft Store application issues.

Steps for Completion

1. What are two ways users can verify internet connectivity?

    a. _____

    _____

2. What command can be used in the Run dialog box to clear the Microsoft Store cache?

    a. _____

3. How can users resolve compatibility conflicts that may affect Microsoft Store functionality?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Troubleshooting Apps
Subtopic: Store App Installation Issues

Objectives covered
5 Management and Troubleshooting
5.2 Troubleshoot operating system and application issues
5.2.5 Resolve Store app installation issues

Repair or Reinstall an Application

Desktop applications in Windows 11 can sometimes experience issues such as crashes, missing features, or slow performance. In such cases, repairing or reinstalling the application can often restore functionality. Understanding when to repair versus reinstall is crucial for effectively resolving software-related problems. Before taking action, users should determine whether the issue is related to the application, a system configuration problem, or a hardware limitation.

Purpose

Upon completing this project, you will better understand how to repair and reinstall desktop applications.

Steps for Completion

1. List at least two signs that an application may need to be repaired or reinstalled.

    a. _____

2. Should users first repair or reinstall an application that is experiencing issues?

    a. _____

3. What can users check if an application is still not working correctly after repairing the application, reinstalling the application, and repairing corrupted system files?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Troubleshooting Apps
Subtopic: Reinstall or Repair Desktop Apps

Objectives covered
5 Management and Troubleshooting
5.2 Troubleshoot operating system and application issues
5.2.6 Reinstall or repair desktop applications

S Mode

Windows 11 in S Mode is designed for security and performance by restricting application installations to those available in the Microsoft Store. S mode is most often used by those who need their devices for personal rather than business reasons. While this mode can enhance system security, it may also create challenges when users need to install software that is not available in the store. Escalating issues related to S Mode require understanding its limitations and the available solutions.

Purpose

Upon completing this project, you will better understand S Mode.

Steps for Completion

1. What are three common reasons for escalating an issue related to S Mode?

    a. _____

2. What are two risks associated with disabling S Mode?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Troubleshooting Apps
Subtopic: S Mode Problem Escalations

Objectives covered
5 Management and Troubleshooting
5.2 Troubleshoot operating system and application issues
5.2.7 Escalate problems dealing with S Mode

LearnKey

Troubleshoot Windows Services

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Troubleshooting Apps
Subtopic: Troubleshoot Services

Objectives covered
5 Management and Troubleshooting
5.2 Troubleshoot operating system and application issues
5.2.8 Troubleshoot services

Windows Services are background processes essential to system functionality, managing everything from network connections to security updates. System performance and application functionality can be affected when services fail to start, stop unexpectedly, or behave abnormally. Troubleshooting service-related issues in Windows 11 requires understanding how to diagnose and fix problems using built-in tools like Services Manager, Event Viewer, and System File Checker.

Purpose

Upon completing this project, you will better understand how to troubleshoot Windows services.

Steps for Completion

1. What should users check in the Services Manager when a service fails to start?

   a. _____

2. Which two error codes can users search for in Event Viewer to find the cause of a service failure?

   a. _____

3. What is the purpose of running System File Checker?

   a. _____

4. What are service dependencies?

   a. _____

Task Manager

Task Manager is an essential utility in Windows 11 that allows users to monitor system performance, manage applications, and troubleshoot unresponsive programs. It provides real-time insights into CPU, memory, disk, and network usage, helping users identify and resolve system slowdowns or application failures. Task Manager can control startup applications, end unresponsive tasks, and manage services effectively.

Purpose

Upon completing this project, you will better understand how to use Task Manager.

Steps for Completion

1. What can cause poor system performance when Windows is starting?

    a. _____

2. Why should users avoid ending critical Windows system processes?

    a. _____

3. What tab in Task Manager has an overview of a system's running and stopped services?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Troubleshooting Apps
Subtopic: Actions with Task Manager

Objectives covered
5 Management and Troubleshooting
5.2 Troubleshoot operating system and application issues
5.2.9 Use Task Manager to disable a startup app, end a task, manage a service

Domain 5

Lesson 4

IT Specialist

Device Configuration and Management

LK LearnKey

Troubleshoot Hardware

Hardware-related issues in Windows 11 often stem from connectivity problems, faulty ports, or power supply failures. Troubleshooting these issues requires a methodical approach, starting with physical connections, checking ports, and verifying power sources.

Purpose

Upon completing this project, you will better understand how to troubleshoot hardware.

Steps for Completion

1. What two issues can cause physical connections for hardware to fail?
    a. _____
    _____
2. How can users troubleshoot a USB connection not working to determine whether the issue is with the device or a specific port?
    a. _____
    _____
3. What are two ways users can troubleshoot power issues on laptops?
    a. _____
    _____
4. What item on desktops can users check to troubleshoot power issues?
    a. _____
    _____
5. What Windows feature can provide further diagnostics for hardware issues?
    a. _____
6. What Windows feature can tell users if a system recognizes a device?
    a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 5
Topic: Hardware and Peripherals
Subtopic: Hardware Troubleshooting Methods

Objectives covered
5 Management and Troubleshooting
5.3 Manage and troubleshoot hardware and peripherals
5.3.1 Hardware troubleshooting methods (connections, ports, power)

Update or Roll Back Drivers

Drivers are essential for ensuring that hardware components such as graphics cards, printers, and network adapters function correctly in Windows 11. Keeping drivers up to date can improve performance and security, while rolling back a driver may be necessary if a new update causes system instability or device malfunctions.

Purpose

Upon completing this project, you will better understand how to update and roll back drivers.

Steps for Completion

1. What Windows feature allows users to update and roll back drivers?

    a. _____

2. What are the two ways users can update drivers?

    a. _____
    _____
    _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Hardware and Peripherals
Subtopic: Update or Roll Back Drivers

Objectives covered
5 Management and Troubleshooting
5.3 Manage and troubleshoot hardware and peripherals
5.3.2 Update or roll back drivers

LK LearnKey

Uninstall and Reinstall Drivers

Uninstalling and reinstalling a device is a common troubleshooting step when dealing with driver-related issues in Windows 11. This process allows Windows to remove potentially corrupted drivers and reinstall fresh, working versions to ensure proper hardware functionality. Whether a device is unresponsive, missing from Device Manager, or displaying errors, reinstalling its drivers can often resolve the issue.

Purpose

Upon completing this project, you will better understand how to uninstall and reinstall drivers.

Steps for Completion

1. What does selecting the Delete the driver software for this device option do?

    a. _____

2. What happens when users restart a computer once a device is uninstalled?

    a. _____

3. When should users download a new driver for a device?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Hardware and Peripherals
Subtopic: Uninstall or Reinstall a Device

Objectives covered
5 Management and Troubleshooting
5.3 Manage and troubleshoot hardware and peripherals
5.3.3 Uninstall or reinstall a device to reconfigure drivers

Device Manager and Disk Management

Device Manager and Disk Management are two critical system tools in Windows 11 that allow users to manage hardware devices and storage configurations. Understanding how to use these tools is essential for troubleshooting hardware issues, updating drivers, and optimizing storage usage.

Purpose

Upon completing this project, you will better understand the purpose of Device Manager and Disk Management.

Steps for Completion

1. What is the purpose of Device Manager?
    a. _____
2. What is the purpose of Disk Management?
    a. _____
3. If a drive is not recognized, it may need to be _____ before it can be used.

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Hardware and Peripherals
Subtopic: Device Manager, Disk Management

Objectives covered
5 Management and Troubleshooting
5.3 Manage and troubleshoot hardware and peripherals
5.3.4 Describe the purpose and capabilities of Device Manager and Disk Management

LearnKey

Peripheral Devices

Peripheral devices such as printers, external hard drives, USB devices, and Bluetooth accessories enhance a computer's functionality. However, connectivity issues can sometimes arise, requiring troubleshooting to ensure proper operation.

Purpose

Upon completing this project, you will better understand how to troubleshoot peripheral devices.

Steps for Completion

1. What first step should users take when a peripheral device is not recognized?

    a. _____

2. What are two ways users can troubleshoot wireless peripheral devices?

    a. _____
    _____
    _____

3. What should users do next if a device is not recognized but the physical connections are intact?

    a. _____

4. How can users refresh a printer's connection if it is not working properly?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Hardware and Peripherals
Subtopic: Troubleshoot Peripherals

Objectives covered
5 Management and Troubleshooting
5.3 Manage and troubleshoot hardware and peripherals
5.3.5 Manage and troubleshoot peripheral device connection

Domain 5

Lesson 5

IT Specialist

Device Configuration and Management

LK LearnKey

Network Connectivity Issues

Network connectivity issues can arise from various factors, including hardware failures, misconfigured settings, or external interference. Effectively troubleshooting these issues requires knowledge of wired (Ethernet) and wireless (Wi-Fi) connections and network diagnostic tools like ping, traceroute, and Nmap.

Purpose

Upon completing this project, you will better understand how to troubleshoot wired and wireless network connections.

Steps for Completion

1. For wired connections, users should ensure that their Ethernet cable is properly connected to the computer and the _____ or switch.
2. What are three items that can weaken wireless connections?
   a. _____
   _____
3. What does the command, ipconfig /release, do?
   a. _____
4. What does the command, ipconfig /renew, do?
   a. _____
   _____
5. What does the command, ipconfig /flushdns, do?
   a. _____
6. What is the purpose of running the tracert command?
   a. _____
   _____
7. What is the purpose of Nmap?
   a. _____

Project Details

Project file
N/A

Estimated completion time
10 minutes

Video reference
Domain 5
Topic: Network and Domain Connections
Subtopic: Wired and Wireless Connections

Objectives covered
5 Management and Troubleshooting
5.4 Manage and troubleshoot device connections to networks and domains
5.4.1 Wired and wireless connections (Ethernet cable, wireless signal strength, SSID, ipconfig options [flushdns, release, renew, all], security key), ping, traceroute, nmap

Join or Remove Devices from a Domain

Joining or removing a device from a domain is an essential task in enterprise environments where organizations use Active Directory (AD) domains to manage users, security policies, and access to network resources. Windows 11 allows administrators and users with the proper permissions to join a device to a domain or remove it when necessary. Understanding how to configure these settings correctly ensures a system functions optimally within an organization's network.

Purpose

Upon completing this project, you will better understand how to join or remove devices from a domain.

Steps for Completion

1. What is the first step when joining a device to a domain?

    a. _____

2. What are two of the most common causes of a device failing to join a domain?

    a. _____

3. What should users ensure is available before removing a device if the device primarily uses a domain account for authentication?

    a. _____

Project Details

Project file
N/A

Estimated completion time
5 minutes

Video reference
Domain 5
Topic: Network and Domain Connections
Subtopic: Remove or Join Devices to Domains

Objectives covered
5 Management and Troubleshooting
5.4 Manage and troubleshoot device connections to networks and domains
5.4.2 Remove or Join devices to domains

Appendix

IT Specialist

Device Configuration and Management

LK LearnKey

Glossary

Domain 1

Term	Definition
Active Directory	A Microsoft service that manages users and resources in a network.
Administrator Account	An account that grants elevated permissions to install software and manage device settings.
Application Shortcuts	Quick links to launch programs or files.
Accelerated Processing Unit (APU)	A processor that combines a CPU and GPU on a single chip to improve performance and energy efficiency.
BIOS	Firmware used to perform hardware initialization and provide runtime services for operating systems.
Control Panel	A legacy interface for configuring system settings in Windows.
Central Processing Unit (CPU)	The primary component of a computer that processes instructions and runs applications.
CUDA	A parallel computing platform and programming model developed by NVIDIA that allows developers to use NVIDIA GPUs for general-purpose processing tasks.
Device Manager	A Windows utility for managing hardware, drivers, and system resources.
Disk Management	A utility used to view and manage disk drives and partitions.
Display Settings	Settings that configure screen resolution, orientation, and multiple monitors.
Domain Administrator	A domain administrator has full control over a Windows domain and can manage users, groups, and settings.
Domain User Account	An account that enables access to network resources managed through Active Directory.
File Explorer	The default file management application in Windows that allows users to browse, open, and organize files.
Graphics Processing Unit (GPU)	A specialized processor that accelerates graphics rendering and parallel computing tasks.
Graphics Card	A component that renders images to a display and may be integrated or dedicated.
Group Policy	Group Policy allows centralized configuration of users and computers within Active Directory.
Group Policy Management Console	A Microsoft administrative tool that provides a unified interface for managing Group Policy Objects (GPOs), allowing administrators to configure and enforce settings across users and computers in an Active Directory environment.
Hard Disk Drive (HDD)	A component that stores digital data using spinning magnetic disks and is often used for bulk storage.
Heatsink	A thermal component that dissipates heat away from electronic components like CPUs.
Local Account	An account that is tied to a single device and lacks centralized management.
Microsoft Account	An account that allows access to Microsoft services and syncs settings across devices.
Motherboard	The main circuit board that connects and manages communication between all computer components.
Multi-Factor Authentication	An authentication method that enhances security by requiring a second form of verification beyond a password.

Term	Definition
PCIe Slot	A high-speed interface on a motherboard used to connect expansion cards like GPUs or network cards.
Power Settings	Settings that manage how the computer uses power and enters sleep or hibernation.
RAM	The temporary memory in a computer which gives applications a place to store and access data on a short-term basis.
Reduced Instruction Set Computing (RISC)	A CPU design model that uses a small, highly optimized set of simple instructions to achieve faster processing and greater efficiency.
Remote Desktop Protocol	A proprietary protocol developed by Microsoft that allows users to connect to and control a remote computer over a network through a graphical interface.
ROCm	An open-source computing platform developed by AMD that enables high-performance, GPU-accelerated computing using AMD GPUs for workloads such as machine learning, scientific computing, and parallel processing.
Settings App	The modern Windows interface for configuring devices, apps, and system preferences.
Solid-State Drive (SSD)	A component that uses flash memory for faster data access and reliability compared to HDDs.
Startup Settings	Settings that allow users to configure how the system behaves on boot, including Safe Mode options.
Storage Device	A device that stores data permanently, such as an HDD or SSD.
System Restore	A tool that allows users to revert system files and settings to a previous state without affecting personal files.
Task Manager	A Windows tool that provides information on running applications, performance, and startup programs.
Taskbar	The taskbar provides access to open windows and pinned applications.
Time Zone Settings	Settings that ensure the system clock aligns with the user's geographical region.
User Profile	A set of settings and data associated with a specific user account.
Virtual Desktop	A tool that allows users to organize open apps across multiple desktops for improved workflow.
Windows Update	A tool that manages operating system patches and updates to keep the system secure and up to date.

Domain 2

Term	Definition
AI	A branch of computer science focused on creating systems or machines that can perform tasks typically requiring human intelligence, such as learning, reasoning, problem-solving, perception, and language understanding.
Credential Manager	A tool that stores usernames and passwords for websites, connected applications, and networks.
Display resolution	The number of distinct pixels displayed on a screen, typically represented as width × height (e.g., 1920×1080), which determines the clarity, detail, and amount of content that can fit on the display.
Group Policy Editor	A Windows administrative tool that allows users to view and modify Group Policy settings, enabling control over user and computer configurations within a local machine or an Active Directory environment.
Hyper-V	A virtualization platform developed by Microsoft that allows users to create and manage virtual machines on Windows systems, enabling multiple operating systems to run concurrently on a single physical host.

Term	Definition
Micosoft Copilot	An AI-powered assistant integrated into Microsoft 365 applications that helps users enhance productivity by generating content, summarizing information, automating tasks, and providing intelligent suggestions based on context.
Microsoft Office	A suite of productivity applications developed by Microsoft, including programs like Word, Excel, PowerPoint, and Outlook, designed to help users create documents, manage data, deliver presentations, and handle communications.
Microsoft Store	A digital distribution platform developed by Microsoft that allows users to download and install apps, games, movies, TV shows, and other content for Windows devices.
Single Sign-On (SSO)	A feature that allows users to sign in once and gain access to multiple related systems without being prompted again.
Windows Registry	A hierarchical database in Windows operating systems that stores low-level settings and configurations for the operating system, installed applications, and hardware devices.
Windows Subsystem for Linux	A compatibility layer developed by Microsoft that allows users to run native Linux command-line tools and applications directly on Windows without the need for a virtual machine or dual-boot setup.
User Account Control (UAC)	A Windows security feature that helps prevent unauthorized changes to the operating system by prompting for permission or administrator credentials before allowing actions that could affect system integrity.

Domain 3

Term	Definition
Cloud computing	The delivery of computing services—including servers, storage, databases, networking, software, and analytics—over the internet ("the cloud") to provide flexible resources, faster innovation, and economies of scale.
Cloud storage	A service model that allows users to save and access data over the internet using remote servers managed by third-party providers, offering scalability, redundancy, and accessibility from any connected device.
Cloud storage services	Online platforms provided by companies like Google, Microsoft, and Amazon, which allow users to store, manage, and access data remotely over the internet, often with features such as file synchronization, sharing, and backup.
Extended File Allocation Table (exFAT)	A file system developed by Microsoft that is optimized for flash storage, offering support for large files and partitions without the compatibility and size limitations of FAT32, and is supported by both Windows and macOS.
File Allocation Table 32 (FAT32)	An older file system format that offers wide compatibility across operating systems and devices, but has limitations such as a maximum file size of 4GB and a partition size of 8TB.
GUID Partition Table (GPT)	A modern partitioning scheme that uses globally unique identifiers to define partitions on a disk, supporting larger storage capacities and more partitions than the older MBR (Master Boot Record) system, and is required for booting drives on UEFI-based systems.
Master Boot Record (MBR)	A legacy partitioning scheme used to define the layout of a disk, supporting up to four primary partitions and a maximum disk size of 2TB, with boot information stored in the first sector of the drive.
New Technology File System (NTFS)	A file system developed by Microsoft for Windows operating systems that supports large file sizes, file permissions, encryption, and advanced data recovery features.

Domain 4

Term	Definition
Antimalware Scan Interface (AMSI)	An interface that allows applications and services to integrate with antivirus software to detect and block malicious scripts.
Audio Settings	Settings that manage input/output devices, volume, and accessibility features like mono audio.
Authenticator apps	Mobile applications that generate time-based one-time passwords (TOTPs) or verification codes used for two-factor authentication (2FA), adding an extra layer of security by requiring users to enter a code from their device in addition to their password.
Biometric authentication	A security process that verifies a person's identity using unique physical or behavioral characteristics—such as fingerprints, facial recognition, or voice patterns—to grant access to devices, systems, or data.
Color Filters	An accessibility tool that assists users with color vision deficiencies by altering on-screen colors.
Contrast Themes	An accessibility tool that increases visual clarity by using high-contrast color combinations.
Firewall	A network security device or software that monitors and filters incoming and outgoing network traffic based on security rules.
Live Captions	An accessibility feature that displays real-time captions for audio content on the screen.
Magnifier	An accessibility tool that enlarges parts of the screen to help users with visual impairments.
Mono Audio	An audio setting that combines stereo sound into one channel for users with hearing loss in one ear.
Mouse Settings	Settings that allow customization of pointer speed, button layout, and scroll behavior.
Narrator	A built-in screen reader that reads aloud interface elements and text for the visually impaired.
On-Screen Keyboard	A virtual keyboard displayed on the screen, useful for accessibility or touchscreen devices.
Password	A secret string of characters used to verify a user's identity and grant access to systems, applications, or data, serving as a fundamental method of authentication and security.
Password Policy	A policy that specifies requirements for user passwords, including length, complexity, and expiration, to improve account security.
Speech Recognition	An accessibility tool that converts spoken language into text and enables voice control of a computer.
Sticky Keys	An accessibility feature that allows modifier keys to remain active after being pressed.
Voice Access	A Windows feature that allows users to control the computer using voice commands.
Windows Defender	A built-in antivirus and antimalware solution for Windows that helps protect your device from threats.

Domain 5

Term	Definition
Account Lockout Policy	The number of failed login attempts allowed before a user account is temporarily locked to prevent brute-force attacks.
BitLocker	A Windows encryption feature that protects data by encrypting entire drives to prevent unauthorized access.
Credential Guard	A security feature in Windows that uses virtualization to protect login credentials from theft.
Drivers	Specialized software programs that enable communication between an operating system and hardware devices, allowing the system to recognize and properly control components like printers, graphics cards, and network adapters.
Microsoft Learn	A free online training platform provided by Microsoft that offers interactive tutorials, modules, and learning paths to help individuals develop skills in Microsoft technologies and prepare for certifications.

Term	Definition
S Mode	A streamlined version of Windows designed for security and performance, allowing only apps from the Microsoft Store and requiring Microsoft Edge for safe browsing to minimize the risk of malware and unauthorized software.
Secure Boot	A UEFI firmware feature that prevents unauthorized operating systems and malware from loading during the startup process.
SmartScreen	A cloud-based security feature in Windows that helps protect users from phishing and malware by screening downloads and websites.
Troubleshooting	The process of identifying, diagnosing, and resolving problems or issues within a system, device, or application to restore normal functionality and performance.

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Objectives

IT Specialist - Device Configuration and Management Objectives	
Domain 1 **Install and Configure Windows**	**Domain 2** **Features, Apps, and Peripherals**
1.1 Recognize basic computer components and explain their functions 1.1.1 RAM 1.1.2 CPU/APU (x86 [32-bit and 64-bit], ARM) 1.1.3 Graphics card 1.1.4 Physical storage 1.1.5 Motherboard	2.1 Manage applications and Windows features 2.1.1 Identify user account requirements and permissions for application installation 2.1.2 Modify application installations 2.1.3 Remove desktop applications 2.1.4 Locate and identify optional Windows features 2.1.5 Describe the purpose of the Microsoft Store 2.1.6 Understand default locations for applications based on architecture 2.1.7 Remote Desktop
1.2 Describe the purpose of Active Directory services 1.2.1 Domain user account 1.2.2 Domain administrator account 1.2.3 Centralized management of users, computers, and groups 1.2.4 Group Policy 1.2.5 Differentiate between local and group security policies and precedence	2.2 Explain the purpose and capabilities of Windows Copilot 2.2.1 Direct users to troubleshooting tools within Windows 11 2.2.2 Can open applications, access Settings, generate content, translate text into different languages, generate programming code and retrieve information 2.2.3 Installed with Windows 11 by default 2.2.4 Limitations of Copilot
1.3 Install Windows using the default settings 1.3.1 Time zone options, Microsoft account vs. local account, upgrade vs. custom install 1.3.2 Installation scenarios include domain-joined and non-domain-joined computers	2.3 Describe the purpose of the Windows Registry 2.3.1 Database of user preferences, application Settings, Windows Settings
1.4 Configure user account options 1.4.1 User account (Microsoft, Domain, or local) 1.4.2 Standard user and administrative account types 1.4.3 User profiles	2.4 Compare and contrast capabilities of peripheral connection types 2.4.1 HDMI (full, mini and micro) 2.4.2 DisplayPort (full, mini) 2.4.3 DVI family of connectors 2.4.4 VGA 2.4.5 USB-A, mini-A, micro-A, 3.0 2.4.6 USB-B, mini-B, micro-B, 3.0 2.4.7 USB-C 2.4.8 Thunderbolt 2.4.9 S/PDIF Optical 2.4.10 Aux audio cable 2.4.11 Converting between the various connection types
1.5 Configure desktop settings 1.5.1 Start menu 1.5.2 Display settings 1.5.3 Application shortcuts 1.5.4 Time zone settings 1.5.5 Taskbar settings 1.5.6 Power settings 1.5.7 Window management (minimize, close, snap)	2.5 Configure projection and display properties 2.5.1 Wireless casting 2.5.2 Orientation 2.5.3 Duplicating vs. extending 2.5.4 Resolution and aspect ratio 2.5.5 ClearType
1.6 Configure accessibility settings to meet users' specific needs 1.6.1 Mouse Settings 1.6.2 Color filters 1.6.3 Contrast themes 1.6.4 Audio Settings	

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IT Specialist - Device Configuration and Management Objectives	
Domain 1 **Install and Configure Windows**	**Domain 2** **Features, Apps, and Peripherals**
1.6.5 Magnifier 1.6.6 Narrator 1.6.7 Sticky Keys 1.6.8 On-screen keyboard 1.6.9 Voice Access 1.6.10 Voice typing (dictation) 1.6.11 Text cursor 1.6.12 Eye control 1.6.13 Live captions	
1.7 Manage updates 1.7.1 Windows Update Settings 1.7.2 Software updates and patches 1.7.3 Optional OS updates 1.7.4 Device driver updates (Microsoft and manufacturer) 1.7.5 Update history	

IT Specialist - Device Configuration and Management Objectives		
Domain 3 **Data Access and Management**	**Domain 4** **Device Security**	**Domain 5** **Management and Troubleshooting**
3.1 Describe cloud services 3.1.1 Cloud storage and collaboration concepts 3.1.2 Identify common cloud storage and service providers, such as Azure, Microsoft 365 (to include but not limited to: SharePoint, OneDrive, Outlook, Teams, Windows 365) 3.1.3 File sharing capabilities and permissions 3.1.4 Offline file synchronization	4.1 Configure Windows Defender Firewall Settings 4.1.1 Allow an application or feature through the Windows Defender Firewall 4.1.2 Compare and contrast private, public, and guest network profiles 4.1.3 Turn firewall off and on (for troubleshooting)	5.1 Perform troubleshooting tasks 5.1.1 Locate and identify Windows troubleshooting tools (such as Event Viewer, Task Manager, Defragment and Optimize Drive) 5.1.2 Gather data to describe issues and support troubleshooting 5.1.3 Research how to remedy issues 5.1.4 Identify when to escalate issues 5.1.5 Force group policy application (gpupdate /force, gpresult) 5.1.6 Recognize that an applied policy could cause a problem
3.2 Describe and configure storage 3.2.1 Understand when and why to use partitioning 3.2.2 Partition and format a drive 3.2.3 Choose a file system to use when formatting the drive (NTFS, FAT32, and exFAT) 3.2.4 Configure File and folder attributes 3.2.5 Identify the effect on permissions and attributes when copying or moving data between file systems 3.2.6 GPT and MBR partition style	4.2 Describe user authentication and configure Windows sign-in options 4.2.1 Multifactor authentication (theory, how it works) 4.2.2 Biometric authentication methods 4.2.3 Windows Hello (Windows 11 only) 4.2.4 Configure Windows sign-in options 4.2.5 What makes a password “strong” 4.2.6 Authenticator apps	5.2 Troubleshoot operating system and application issues 5.2.1 Reset or roll back the operating system 5.2.2 Advanced startup (System repair) 5.2.3 Features of safe mode 5.2.4 Use troubleshooting tools to identify application compatibility issues 5.2.5 Resolve Store app installation issues 5.2.6 Reinstall or repair desktop applications 5.2.7 Escalate problems dealing with ‘S’ mode 5.2.8 Troubleshoot services 5.2.9 Use Task Manager to disable a startup app, end a task, manage a service
3.3 Describe and configure local and network file sharing and permissions 3.3.1 File and share permissions 3.3.2 Effective permissions	4.3 Describe various security threats 4.3.1 Computer viruses (worms, trojan horse, logic bombs, ransomware) 4.3.2 Adware	5.3 Manage and troubleshoot hardware and peripherals 5.3.1 Hardware troubleshooting methods (connections, ports, power)

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IT Specialist - Device Configuration and Management Objectives		
Domain 3 Data Access and Management	**Domain 4 Device Security**	**Domain 5 Management and Troubleshooting**
3.3.3 Basic and advanced permissions 3.3.4 Public, basic, and advanced shares 3.3.5 Map drives 3.3.6 Describe taking ownership of files or folders	4.3.3 Spyware 4.3.4 Denial of Service (DoS) attacks 4.3.5 Social engineering attacks (to include, but not limited to Phishing, smishing, vishing, dumpster diving, spoofing, and clone phishing) 4.3.6 Physical attacks (errant thumb drives, theft, shoulder surfing, screen scrapers)	5.3.2 Update or roll back drivers 5.3.3 Uninstall or reinstall a device to reconfigure drivers 5.3.4 Describe the purpose and capabilities of Device Manager and Disk Management 5.3.5 Manage and troubleshoot peripheral device connection
3.4 Manage backup and restore of user files and states 3.4.1 Set file versioning/history Settings 3.4.2 Perform a full disk backup to the cloud 3.4.3 Perform a full disk restore 3.4.4 Restore previous versions of files	4.4 Describe how to respond to various malware and social engineering attacks 4.4.1 Computer viruses 4.4.2 Adware 4.4.3 Spyware 4.4.4 Phishing 4.4.5 Physical attacks (errant thumb drives) 4.4.6 Antimalware program configuration options 4.4.7 Analyze Antimalware scan results	5.4 Manage and troubleshoot device connections to networks and domains 5.4.1 Wired and wireless connections (Ethernet cable, wireless signal strength, SSID, ipconfig options [flushdns, release, renew, all], security key), ping, traceroute, nmap 5.4.2 Remove or Join devices to domains
	4.5 Manage User Account Control (UAC) Settings 4.5.1 Describe the function of UAC 4.5.2 Identify appropriate UAC Settings for specific purposes 4.5.3 Elevate permissions in UAC	

Lesson Plan

Approximately 29.5 hours of video, labs, and projects

IT Specialist

Device Configuration and Management

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Domain 1 Lesson Plan

Domain 1 - Install and Configure Windows [approximately 9 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	Install and Configure Windows: Pre-Assessment			
Lesson 1 Video time - 00:17:50 Exercise Lab time - 00:08:00 Workbook time - 00:30:00	**Basic Computer Components** How to Study for This Exam RAM CPU, APU, ARM Graphics Card Physical Storage Motherboard	1.1 Recognize basic computer components and explain their functions 1.1.1 RAM 1.1.2 CPU/APU (x86 [32-bit and 64-bit], ARM) 1.1.3 Graphics Card 1.1.4 Physical Storage 1.1.5 Motherboard	DDR RAM CPUs and APUs	RAM – pg. 8 N/A CPU, APU, ARM – pg. 9 N/A Graphics Card – pg. 10 N/A Physical Storage – pg. 11 N/A Motherboard – pg. 12 N/A
Lesson 2 Video time - 00:12:20 Exercise Lab time - 00:04:00 Workbook time - 00:25:00	**Active Directory Services** Domain User Account Domain Administrator Account Centralized Management Group Policy Local and Group Security Policies	1.2 Describe the purpose of Active Directory services 1.2.1 Domain User Account 1.2.2 Domain administrator account 1.2.3 Centralized Management of users, computers, and groups 1.2.4 Group Policy 1.2.5 Differentiate between local and group security policies and precedence	Security Policy Types	Domain User Account – pg. 14 N/A Domain Administrator Account – pg. 15 N/A Centralized Management – pg. 16 N/A Group Policy – pg. 17 N/A Local and Group Security Policies – pg. 18 N/A
Lesson 3 Video time - 00:06:08 Exercise Lab time - 00:04:00 Workbook time - 00:15:00	**Install Windows** Installation and Account Options Domain and Non-Domain Devices	1.3 Install Windows using the default settings 1.3.1 Time zone options, Microsoft account vs. local account, upgrade vs. custom install 1.3.2 Installation scenarios include domain-joined and non-domain-joined computers	Time Zone Settings	Installation and Account Options – pg. 20 N/A Domain and Non-Domain Devices – pg. 21 N/A
Lesson 4 Video time - 00:09:23 Exercise Lab time - 00:04:00 Workbook time - 00:15:00	**User Account Types** Types of User Accounts User and Administrative Accounts User Profiles	1.4 Configure user account options 1.4.1 User account (Microsoft, Domain, or local) 1.4.2 Standard user and administrative account types 1.4.3 User profiles	Standard User Account	User Accounts Types – pg. 23 N/A User and Administrative Accounts – pg. 24 N/A User Profiles – pg. 25 N/A
Lesson 5 Video time - 00:16:46 Exercise Lab time - 00:08:00	**Desktop Settings** Start Menu Display Settings Application Shortcuts Time Zone Settings	1.5 Configure desktop settings 1.5.1 Start menu 1.5.2 Display settings 1.5.3 Application shortcuts 1.5.4 Time zone settings	Customize the Start Menu Create a Shortcut	Start Menu – pg. 27 N/A Display Settings – pg. 28 N/A Application Shortcuts – pg. 29 N/A

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Domain 1 - Install and Configure Windows [approximately 9 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Workbook time - 00:40:00				Time Zone Settings – pg. 30 N/A
Lesson 6 Video time - 00:10:39 Exercise Lab time - 00:12:00 Workbook time - 00:25:00	**Taskbar and Power Settings** Taskbar Settings Taskbar Personalization Power Settings Windows Management	1.5 Configure desktop settings 1.5.5 Taskbar settings 1.5.6 Power settings 1.5.7 Window management (minimize, close, snap)	Taskbar Alignment Pin and Unpin Close a Window	Taskbar Personalization – pg. 32 N/A Power Settings – pg. 33 N/A Windows Management – pg. 34 N/A
Lesson 7 Video time - 00:14:08 Exercise Lab time - 00:12:00 Workbook time - 00:35:00	**Accessibility Settings** Mouse Settings Mouse Settings in the Control Panel Color Filters Contrast Themes Audio Settings Magnifier Narrator	1.6 Configure accessibility settings to meet users' specific needs 1.6.1 Mouse setting 1.6.2 Color filters 1.6.3 Contrast themes 1.6.4 Audio settings 1.6.5 Magnifier 1.6.6 Narrator	Mouse Keys Color Filters Contrast Themes	Mouse Settings – pg. 36 N/A Color Filters and Contrast Themes – pg. 37 N/A Audio Settings – pg. 38 N/A Magnifier – pg. 39 N/A Narrator – pg. 40 N/A
Lesson 8 Video time - 00:13:06 Exercise Lab time - 00:00:00 Workbook time - 00:35:00	**Keyboard, Voice, and Cursor Settings** Sticky Keys On-Screen Keyboard Voice Access Voice Typing Text Cursor Eye Control Live Captions	1.6 Configure accessibility settings to meet users' specific needs 1.6.7 Sticky keys 1.6.8 On-screen keyboard 1.6.9 Voice Access 1.6.10 Voice Typing (Dictation) 1.6.11 Text cursor 1.6.12 Eye control 1.6.13 Live captions	N/A	Sticky Keys – pg. 42 N/A On-Screen Keyboard – pg. 43 N/A Voice Access and Voice Typing – pg. 44 N/A Text Cursor and Eye Control – pg. 45 N/A Live Captions – pg. 46 N/A
Lesson 9 Video time - 00:11:14 Exercise Lab time - 00:08:00 Workbook time - 00:35:00	**OS and App Updates** Windows Update Settings Software Updates and Patches Optional OS Updates Device Driver Updates Update History	1.7 Manage updates 1.7.1 Windows Update settings 1.7.2 Software updates and patches 1.7.3 Optional OS updates 1.7.4 Device driver updates (Microsoft and manufacturer) 1.7.5 Update history	Microsoft Store Updates Optional Windows Updates	Windows Update Settings – pg. 48 N/A Software Updates and Patches – pg. 49 N/A Optional OS Updates – pg. 50 N/A Device Driver Updates – pg. 51 N/A Update History – pg. 52 N/A
Post-Assessment Assessment time - 01:00:00	Install and Configure Windows: Post-Assessment			

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Domain 2 Lesson Plan

Domain 2 - Features, Apps, and Peripherals [approximately 6 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	Features, Apps, and Peripherals: Pre-Assessment			
Lesson 1 Video time - 00:09:54 Exercise Lab time - 00:12:00 Workbook time - 00:15:00	**Manage Apps** User Account Requirements Modify App Installations Remove Desktop Apps	2.1 Manage applications and Windows features 2.1.1 Identify user account requirements and permissions for application installation 2.1.2 Modify application installations 2.1.3 Remove desktop applications	View Installed Apps Use the Microsoft Store Programs and Features	Install Applications – pg. 54 N/A Modify Applications – pg. 55 N/A Remove Desktop Applications – pg. 56 N/A
Lesson 2 Video time - 00:08:50 Exercise Lab time - 00:04:00 Workbook time - 00:20:00	**Manage Windows Features** Optional Windows Features Microsoft Store Purpose App Default Locations Remote Desktop	2.1 Manage applications and Windows features 2.1.4 Locate and identify optional Windows features 2.1.5 Describe the purpose of the Microsoft Store 2.1.6 Understand default locations for applications based on architecture 2.1.7 Remote Desktop	Remote Desktop Connection	Optional Features – pg. 58 N/A The Microsoft Store – pg. 59 N/A Application Storage – pg. 60 N/A Remote Desktop – pg. 61 N/A
Lesson 3 Video time - 00:10:06 Exercise Lab time - 00:00:00 Workbook time - 00:20:00	**Windows Copilot** Troubleshooting Directions Uses for Copilot Default Installation Limitations of Copilot	2.2 Explain the purpose and capabilities of Windows Copilot 2.2.1 Direct users to troubleshooting tools within Windows 11 2.2.2 Can open applications, access Settings, generate content, translate text into different languages, generate programming code and retrieve information 2.2.3 Installed with Windows 11 by default 2.2.4 Limitations of Copilot	N/A	Using Copilot – pg. 63 N/A Copilot Limitations – pg. 64 N/A
Lesson 4 Video time - 00:11:41 Exercise Lab time - 00:00:00 Workbook time - 00:25:00	**Windows Registry** Preferences and Settings **Video Connection Types** HDMI DisplayPort DVI	2.3 Describe the purpose of the Windows Registry 2.3.1 Database of user preferences, application Settings, Windows Settings 2.4 Compare and contrast capabilities of peripheral connection types 2.4.1 HDMI (full, mini and micro) 2.4.2 DisplayPort (full, mini) 2.4.3 DVI family of connectors	N/A	The Windows Registry – pg. 66 N/A HDMI – pg. 67 N/A DisplayPort – pg. 68 N/A DVI – pg. 69 N/A

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Domain 2 - Features, Apps, and Peripherals [approximately 6 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Lesson 5 Video time - 00:08:31 Exercise Lab time - 00:00:00 Workbook time - 00:30:00	**VGA, USB, and Thunderbolt** VGA USB-A USB-B USB-C Thunderbolt	2.4 Compare and contrast capabilities of peripheral connection types 2.4.4 VGA 2.4.5 USB-A, mini-A, micro-A, 3.0 2.4.6 USB-B, mini-B, micro-B, 3.0 2.4.7 USB-C 2.4.8 Thunderbolt	USB-A	VGA – pg. 71 N/A USB-A, Mini-A, Micro-A, and USB 3.0 – pg. 72 N/A USB-B, Mini-B, Micro-B, and USB 3.0 – pg. 73 N/A USB-C – pg. 74 N/A Thunderbolt – pg. 75 N/A
Lesson 6 Video time - 00:06:57 Exercise Lab time - 00:00:00 Workbook time - 00:20:00	**Audio and Conversion Connectors** S/PDIF Optical AUX Audio Cable Connection Type Conversions	2.4 Compare and contrast capabilities of peripheral connection types 2.4.9 S/PDIF Optical 2.4.10 Aux audio cable 2.4.11 Converting between the various connection types	N/A	S/PDIF – pg. 77 N/A AUX Cable – pg. 78 N/A HDMI Converters – pg. 79 N/A
Lesson 7 Video time - 00:11:17 Exercise Lab time - 00:00:00 Workbook time - 00:40:00	**Projection and Display Properties** Wireless Casting Orientation Duplicating vs. Extending Resolution and Aspect Ratio ClearType	2.5 Configure projection and display properties 2.5.1 Wireless casting 2.5.2 Orientation 2.5.3 Duplicating vs. extending 2.5.4 Resolution and aspect ratio 2.5.5 ClearType	N/A	Wireless Casting – pg. 81 N/A Display Orientation – pg. 82 N/A Duplicating and Extending - pg. 83 N/A Resolution and Aspect Ratio – pg. 84 N/A ClearType – pg. 85 N/A
Post-Assessment Assessment time - 01:00:00	Features, Apps, and Peripherals: Post-Assessment			

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Domain 3 Lesson Plan

Domain 3 - Data Access and Management [approximately 4 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	Data Access and Management: Pre-Assessment			
Lesson 1 Video time - 00:08:23 Exercise Lab time - 00:04:00 Workbook time - 00:20:00	**Cloud Services** Cloud Storage and Collaboration Storage and Service Providers File Sharing Offline File Synchronization	3.1 Describe cloud services 3.1.1 Cloud storage and collaboration concepts 3.1.2 Identify common cloud storage and service providers, such as Azure, Microsoft 365 (to include but not limited to: SharePoint, OneDrive, Outlook, Teams, Windows 365) 3.1.3 File sharing capabilities and permissions 3.1.4 Offline file synchronization	File Sharing	Cloud Services – pg. 87 N/A Storage and Service Providers – pg. 88 N/A File Sharing and Syncronization – pg. 89 N/A
Lesson 2 Video time - 00:06:52 Exercise Lab time - 00:08:00 Workbook time - 00:30:00	**Storage** Partition and Format a Drive File and Folder Attributes Moving and Copying GPT and MBR Partition Style	3.2 Describe and configure storage 3.2.1 Understand when and why to use partitioning 3.2.2 Partition and format a drive 3.2.3 Choose a file system to use when formatting the drive (NTFS, FAT32, and exFAT) 3.2.4 Configure File and folder attributes 3.2.5 Identify the effect on permissions and attributes when copying or moving data between file systems 3.2.6 GPT and MBR partition style	Attributes GPT and MBR	Partition and Format a Drive – pg. 91 N/A Files and Folders – pg. 92 N/A GPT and MBR Partition Styles – pg. 93 N/A
Lesson 3 Video time - 00:07:55 Exercise Lab time - 00:12:00 Workbook time - 00:25:00	**Local and Network File Permissions** File and Share Permissions Effective Permissions Basic and Advanced Permissions Public, Basic, and Advanced Shares Map Drives Taking Ownership	3.3 Describe and configure local and network file sharing and permissions 3.3.1 File and share permissions 3.3.2 Effective permissions 3.3.3 Basic and advanced permissions 3.3.4 Public, basic, and advanced shares 3.3.5 Map drives 3.3.6 Describe taking ownership of files or folders	View Effective Permissions Public, Basic, and Advanced Shares Map a Network Drive Taking Ownership	File, Share, and Effective Permissions – pg. 95 N/A Basic and Advanced Permissions and Shares – pg. 96 N/A Mapping a Drive and Taking Ownership – pg. 97 N/A
Lesson 4 Video time - 00:04:55 Exercise Lab time - 00:04:00 Workbook time - 00:10:00	**Backup and Restore** File Versioning and History Full Disk Backup to the Cloud Full Disk Restore Restore Previous Versions of Files	3.4 Manage backup and restore of user files and states 3.4.1 Set file versioning/history Settings 3.4.2 Perform a full disk backup to the cloud 3.4.3 Perform a full disk restore 3.4.4 Restore previous versions of files	File History	Backup and Restore Files – pg. 99 N/A
Post-Assessment Assessment time - 01:00:00	Data Access and Management Post-Assessment			

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Domain 4 Lesson Plan

Domain 4 - Device Security [approximately 4.5 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	Device Security: Pre-Assessment			
Lesson 1 Video time - 00:05:20 Exercise Lab time - 00:04:00 Workbook time - 00:15:00	**Windows Defender Firewall** Allow an Application or Feature Network Profiles Firewall Troubleshooting	4.1 Configure Windows Defender Firewall settings 4.1.1 Allow an application or feature through the Windows Defender Firewall 4.1.2 Compare and contrast private, public, and guest network profiles 4.1.3 Turn firewall off and on (for troubleshooting)	Allow an Application	Allow an App or Feature – pg. 101 N/A Network Profiles – pg. 102 N/A Firewall Troubleshooting – pg. 103 N/A
Lesson 2 Video time - 00:08:49 Exercise Lab time - 00:00:00 Workbook time - 00:25:00	**User Authentication Options** Multifactor Authentication Biometric Authentication Methods Windows Hello Configure Windows Sign-In Options Strong Passwords Authenticator Apps	4.2 Describe user authentication and configure Windows sign-in options 4.2.1 Multifactor authentication (theory, how it works) 4.2.2 Biometric authentication methods 4.2.3 Windows Hello (Windows 11 only) 4.2.4 Configure Windows sign-in options 4.2.5 What makes a password “strong” 4.2.6 Authenticator apps	N/A	Multifactor Authentication – pg. 105 N/A Biometric Authentication – pg. 106 N/A Windows Sign-In Options – pg. 107 N/A Strong Passwords – pg. 108 N/A Authenticator Apps – pg. 109 N/A
Lesson 3 Video time - 00:15:31 Exercise Lab time - 00:00:00 Workbook time - 00:40:00	**Various Security Threats** Computer Viruses Adware Spyware Denial of Service (DoS) Attacks Social Engineering Attacks Physical Attacks	4.3 Describe various security threats 4.3.1 Computer viruses (worms, trojan horse, logic bombs, ransomware) 4.3.2 Adware 4.3.3 Spyware 4.3.4 Denial of Service (DoS) attacks 4.3.5 Social engineering attacks (to include, but not limited to Phishing, smishing, vishing, dumpster diving, spoofing, and clone phishing) 4.3.6 Physical attacks (errant thumb drives, theft, shoulder surfing, screen scrapers) 4.4 Describe how to respond to various malware and social engineering attacks 4.4.1 Computer viruses 4.4.2 Adware 4.4.3 Spyware 4.4.4 Phishing 4.4.5 Physical attacks (errant thumb drives)	N/A	Computer Viruses – pg. 111 N/A Adware – pg. 112 N/A Spyware – pg. 113 N/A Denial of Service (DoS) Attacks – pg. 114 N/A Social Engineering Attacks – pg. 115 N/A Physical Attacks – pg. 116 N/A

Domain 4 - Device Security [approximately 4.5 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Lesson 4 Video time - 00:07:52 Exercise Lab time - 00:00:00 Workbook time - 00:15:00	**Antimalware Options and Results** Antimalware App Options Analyze Antimalware Scan Results	4.4 Describe how to respond to various malware and social engineering attacks 4.4.6 Antimalware program configuration options 4.4.7 Analyze Antimalware scan results	N/A	Antimalware App Options – pg. 118 N/A Antimalware Scan Results – pg. 119 N/A
Lesson 5 Video time - 00:10:43 Exercise Lab time - 00:08:00 Workbook time - 00:10:00	**UAC Settings** Describe the Function of UAC UAC Setting Levels Elevate Permissions in UAC	4.5 Manage User Account Control (UAC) settings 4.5.1 Describe the function of UAC 4.5.2 Identify appropriate UAC settings for specific purposes 4.5.3 Elevate permissions in UAC	User Account Control Require Elevated Permissions	User Account Control (UAC) – pg. 121 N/A
Post-Assessment Assessment time - 01:00:00	Device Security: Post-Assessment			

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Domain 5 Lesson Plan

Domain 5 - Management and Troubleshooting [approximately 6 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
Pre-Assessment Assessment time - 00:30:00	Management and Troubleshooting: Pre-Assessment			
Lesson 1 Video time - 00:17:47 Exercise Lab time - 00:12:00 Workbook time - 00:35:00	**Troubleshooting Tasks** Windows Troubleshooting Tools Gather Data on Issues Research Issue Remedies Identify When to Escalate Issues Force Group Policy Application Problems with Applied Policies	5.1 Perform troubleshooting tasks 5.1.1 Locate and identify Windows troubleshooting tools (such as event viewer, task manager, defragment and optimize drive) 5.1.2 Gather data to describe issues and support troubleshooting 5.1.3 Research how to remedy issues 5.1.4 Identify when to escalate issues 5.1.5 Force group policy application (gpupdate /force, gpresult) 5.1.6 Recognize that an applied policy could cause a problem	Optimize Windows PowerShell Force a Group Policy Update	Windows Troubleshooting Tools – pg. 123 N/A Gather Data on Issues – pg. 124 N/A Research Issue Remedies – pg. 125 N/A Identify When to Escalate Issues – pg. 126 N/A Force Group Policy Application – pg. 127 N/A Problems with Applied Policies – pg. 128 N/A
Lesson 2 Video time - 00:14:27 Exercise Lab time - 00:08:00 Workbook time - 00:25:00	**Operating Systems and Applications** Resets and Rollbacks Advanced Startup Safe Mode Features Identify App Compatibility Issues	5.2 Troubleshoot operating system and application issues 5.2.1 Reset or roll back the operating system 5.2.2 Advanced startup (System repair) 5.2.3 Features of safe mode 5.2.4 Use troubleshooting tools to identify application compatibility issues	Compatability Troubleshooter	Resets and Rollbacks – pg. 130 N/A Advanced Startup – pg. 131 N/A Safe Mode – pg. 132 N/A Application Compatibility Issues – pg. 133 N/A
Lesson 3 Video time - 00:16:13 Exercise Lab time - 00:12:00 Workbook time - 00:30:00	**Troubleshooting Apps** Store App Installation Issues Reinstall or Repair Desktop Apps S Mode Problem Escalations Troubleshoot Services Actions with Task Manager	5.2 Troubleshoot operating system and application issues 5.2.5 Resolve Store app installation issues 5.2.6 Reinstall or repair desktop applications 5.2.7 Escalate problems dealing with 'S' mode 5.2.8 Troubleshoot services 5.2.9 Use Task Manager to disable a startup app, end a task, manage a service	Troubleshooting Apps Troubleshoot Services Disable Startup Applications	Microsoft Store Application Issues – pg. 135 N/A Repair or Reinstall an Application – pg. 136 N/A S Mode – pg. 137 N/A Troubleshoot Windows Services –

Domain 5 - Management and Troubleshooting [approximately 6 hours of videos, labs, and projects]				
Lesson	**Lesson Topic and Subtopics**	**Objectives**	**Exercise Labs**	**Workbook Projects and Files**
				pg. 138 N/A Task Manager – pg. 139 N/A
Lesson 4 Video time - 00:12:01 Exercise Lab time - 00:04:00 Workbook time - 00:30:00	**Hardware and Peripherals** Hardware Troubleshooting Methods Update or Roll Back drivers Uninstall or Reinstall a Device Device Manager, Disk Management Troubleshoot Peripherals	5.3 Manage and troubleshoot hardware and peripherals 5.3.1 Hardware troubleshooting methods (connections, ports, power) 5.3.2 Update or roll back drivers 5.3.3 Uninstall or reinstall a device to reconfigure drivers 5.3.4 Describe the purpose and capabilities of Device Manager and Disk Management 5.3.5 Manage and troubleshoot peripheral device connection	Update Drivers	Troubleshoot Hardware – pg. 141 N/A Update or Roll Back Drivers – pg. 142 N/A Uninstall and Reinstall Drivers – pg. 143 N/A Device Manager and Disk Management – pg. 144 N/A Peripheral Devices – pg. 145 N/A
Lesson 5 Video time - 00:08:12 Exercise Lab time - 00:04:00 Workbook time - 00:15:00	**Network and Domain Connections** Wired and Wireless Connections Remove or Join Devices to Domains	5.4 Manage and troubleshoot device connections to networks and domains 5.4.1 Wired and wireless connections (Ethernet cable, wireless signal strength, SSID, ipconfig options [flushdns, release, renew, all], security key), ping, traceroute, nmap 5.4.2 Remove or Join devices to domains	Troubleshooting Network Connectivity	Network Connectivity Issues – pg. 147 N/A Join or Remove Devices from a Domain – pg. 148 N/A
Post-Assessment Assessment time - 01:00:00	Management and Troubleshooting: Post-Assessment			